

Cardio-Diffusion: Generative Reconstruction of ECG Signals Using Diffusion Models for Arrhythmia Classification

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Abstract

Continuous ambulatory Holter monitoring is essential for detecting intermittent arrhythmias; however, signal loss caused by electrode disconnection, motion artifacts, or poor electrode contact can make clinically relevant ECG segments non-diagnostic. This work presents **Cardio-Diffusion**, an integrated framework for reconstructing missing electrocardiographic (ECG) segments using conditional denoising diffusion probabilistic models and evaluating their usefulness for downstream arrhythmia analysis. The proposed system combines ECG preprocessing, artificial missing-segment simulation, a mask-guided conditional 1D U-Net with attention, anatomical reconstruction evaluation, and binary Normal-versus-Arrhythmia classification. Experiments were performed on the MIT-BIH Arrhythmia Database using two-lead ECG windows of 512 samples. Reconstruction quality was evaluated only inside the hidden region using PRD, MSE, MAE, RMSE, and Pearson correlation. Cardio-Diffusion V4 achieved a PRD of 6.38% in the ST segment and 11.88% in the QRS complex, showing that smoother ECG regions are reconstructed with lower error, while abrupt high-frequency morphology remains more challenging. In the reconstruction-based diagnostic evaluation, the classifier achieved 84.5% accuracy and approximately 95.1% arrhythmia sensitivity, correctly identifying 117 of 123 arrhythmic samples. As a secondary benchmark on original ECG signals, a beat-level classifier achieved 98.99% accuracy and 98.50% F1-score using a random beat-level split; this result is reported as morphological separability rather than patient-independent clinical validation. Overall, Cardio-Diffusion demonstrates the feasibility of using conditional diffusion models as a signal-rescue strategy for incomplete ECG recordings, while highlighting the need for future validation with real Holter artifacts, strict patient-level splits, and larger multi-lead datasets.

Keywords: ECG reconstruction; diffusion model; DDPM; signal imputation; arrhythmia classification; Holter monitoring; MIT-BIH; biomedical signal processing; generative models; missing data

1. Introduction

Cardiovascular diseases remain one of the leading causes of morbidity and mortality worldwide, and cardiac arrhythmias represent a major diagnostic challenge because many clinically relevant rhythm abnormalities occur intermittently. Paroxysmal arrhythmias may not appear during a conventional resting electrocardiogram (ECG), making continuous ambulatory monitoring essential for their detection. Holter monitoring systems, typically used over 24 to 48 hours, record the electrical activity of the heart during daily activities and provide a broader temporal window for identifying abnormal rhythm events [1,2].

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Despite their clinical value, Holter recordings are frequently affected by technical problems in real-world conditions. Patient movement, sweating, poor electrode adhesion, cable motion, and intermittent electrode detachment can introduce corrupted intervals or complete gaps in the ECG signal. These interruptions may affect diagnostically important regions such as the QRS complex, ST segment, or T wave. When this occurs, part of the recording may become clinically non-diagnostic, reducing diagnostic confidence and potentially requiring repeated monitoring. Signal-quality assessment and artifact detection methods have been proposed to identify unreliable ECG segments, but most of them focus on detecting or rejecting corrupted intervals rather than reconstructing the missing morphology [6,8].

Traditional strategies for handling missing ECG samples are usually based on linear interpolation, polynomial interpolation, spline interpolation, wavelet-based denoising, or smoothing filters. Although these methods can generate continuous mathematical curves, they do not explicitly model the physiological morphology of ECG signals. As a result, they may fail to reproduce abrupt QRS slopes, subtle P waves, or clinically relevant ST-T variations. This limitation is especially important when the missing interval overlaps with the QRS complex, because small distortions in this region may affect beat detection, rhythm interpretation, and arrhythmia classification [4,5,7]. Therefore, there is a need for reconstruction methods that go beyond smooth mathematical filling and instead learn the underlying morphology of ECG signals.

Deep learning has substantially improved automatic ECG analysis. Convolutional neural networks, residual architectures, attention mechanisms, recurrent models, and Transformer-based models have achieved strong performance in arrhythmia classification and ECG interpretation when complete and clean ECG segments are available [10–13,15]. However, classification models alone do not solve the problem of missing diagnostic information. If a relevant ECG segment is corrupted or absent, the classifier receives incomplete morphology and may produce unreliable predictions. This is particularly problematic in ambulatory monitoring, where signal loss can affect exactly the regions needed for rhythm classification.

Generative models offer a possible solution by learning the distribution of ECG morphology and using this knowledge to synthesize or reconstruct missing waveform segments. Generative Adversarial Networks (GANs) have been explored for ECG synthesis and data augmentation, but adversarial training may suffer from instability, mode collapse, and limited control over the generated morphology [17–19]. These limitations are relevant in biomedical signals because a waveform may appear visually plausible while failing to preserve clinically meaningful structure.

Denoising Diffusion Probabilistic Models (DDPMs) have recently emerged as a stable alternative for generative modeling. Diffusion models learn a gradual denoising process by progressively adding Gaussian noise to the data and training a neural network to reverse this process [22–24]. In time-series applications, conditional score-based diffusion models have shown strong performance for probabilistic imputation by conditioning the generation process on observed values [26,27]. In the ECG domain, diffusion-based approaches have been adapted for signal synthesis, lead reconstruction, biomedical signal completion, and morphology recovery [29–35].

However, many existing diffusion-based ECG approaches focus on generating isolated beats, synthesizing complete ECG signals, reconstructing missing leads, or imputing generic time-series values. Fewer studies directly address the practical problem of reconstructing local missing intervals inside ECG windows while evaluating whether the reconstructed signal remains useful for downstream arrhythmia classification. This distinc-

tion is important because a reconstructed ECG segment should not only look plausible; it should also preserve diagnostic characteristics relevant to rhythm analysis.

To address this gap, this work proposes **Cardio-Diffusion**, an integrated framework for ECG missing-segment reconstruction and arrhythmia analysis. The proposed system combines digital ECG preprocessing, artificial missing-segment simulation, conditional DDPM-based reconstruction, signal-level evaluation, anatomical-region analysis, and binary Normal-versus-Arrhythmia classification. The generative model is implemented as a conditional 1D U-Net with attention and mask conditioning. The binary mask explicitly informs the network which samples are visible and which samples must be reconstructed, allowing the model to focus on the hidden region while using the surrounding ECG context.

The general objective of this work is to design and evaluate a generative reconstruction framework capable of imputing missing ECG segments while preserving clinically relevant morphology. The specific objectives are: (1) to implement an ECG preprocessing pipeline based on filtering, segmentation, and normalization; (2) to adapt a conditional diffusion model for mask-guided ECG reconstruction; (3) to evaluate reconstruction quality using quantitative metrics computed only inside the hidden region; and (4) to assess whether reconstructed ECG segments preserve diagnostic information for binary arrhythmia classification.

The main contributions of this work are summarized as follows:

- A conditional DDPM framework is adapted for local missing-segment reconstruction in two-lead ECG windows.
- A binary mask-guided input representation is used to explicitly distinguish known and hidden ECG samples.
- Reconstruction quality is evaluated using PRD, MSE, MAE, RMSE, and Pearson correlation computed only inside the hidden region.
- Anatomical-region reconstruction is analyzed separately for smoother and more complex ECG components, including ST/T regions and QRS morphology.
- The reconstructed signals are evaluated through a downstream binary arrhythmia classifier to assess diagnostic usefulness.
- A secondary beat-level classifier on original ECG signals is reported as a morphological separability benchmark, while explicitly acknowledging the limitation of random beat-level splitting.

1.1. Related Work

The development of ECG processing methods has evolved from classical filtering and artifact detection techniques to modern deep learning and generative modeling approaches. Early ECG processing methods focused mainly on noise reduction, beat detection, and waveform delineation. Artifact detection and signal-quality assessment systems can identify unreliable segments in ambulatory ECG recordings, but they generally do not reconstruct the missing or discarded signal [6,8]. Similarly, ECG delineation algorithms can identify relevant waveform components such as P waves, QRS complexes, and T waves, but they typically assume that the signal is already complete [5,51].

Deep learning classifiers have improved automatic arrhythmia recognition using complete ECG inputs. Patient-specific 1D CNNs, deep convolutional architectures, residual networks, recurrence-plot representations, attention-based temporal convolutional networks, and Transformer-based models have all been explored for ECG classification [10–16]. Nevertheless, these methods are primarily designed for classification, not for recovering missing ECG morphology. Their performance can degrade when diagnostically relevant waveform components are corrupted or absent.

Generative ECG methods were introduced to synthesize beats, augment imbalanced datasets, and reconstruct missing signal components. GAN-based approaches, including patient-specific GANs, have shown potential for ECG synthesis and classification improvement [17–21]. However, adversarial models may be difficult to train and can generate unstable or insufficiently diverse morphologies. These limitations have motivated the exploration of diffusion models for ECG synthesis and imputation.

Diffusion models provide a stable generative framework by learning to reverse a gradual noising process [22–24]. In time-series imputation, CSDI demonstrated that conditional score-based diffusion can reconstruct missing values by conditioning the generation process on observed data [27]. ECG-specific diffusion methods have extended this idea to ECG synthesis, imputation, lead reconstruction, and morphology recovery. SSSD-ECG incorporated structured state-space models into diffusion-based ECG generation [29]. DiffECG explored probabilistic ECG synthesis [30]. PulseDiff used physiological template priors for ECG imputation [31]. BeatDiff and RhythmDiff investigated diffusion-based ECG morphology reconstruction from indirect or incomplete signals [32,33]. Other recent works have explored latent diffusion, biomedical signal completion, and clinically informed imputation [34–37].

Although these studies demonstrate the promise of diffusion models for biomedical signal reconstruction, several gaps remain. Some methods focus on isolated beat synthesis rather than local gap reconstruction. Others reconstruct full leads or long ECG sequences but do not evaluate the diagnostic usefulness of the reconstructed segments. In contrast, Cardio-Diffusion focuses on local missing-segment reconstruction inside ECG windows and evaluates the reconstructed signals using both signal-level metrics and downstream arrhythmia classification. Table 1 summarizes representative studies related to ECG processing, synthesis, imputation, and diffusion-based reconstruction.

Table 1. Comparison of representative studies in ECG processing, synthesis, imputation, and reconstruction.

Author (Year)	Model	Method	Dataset	Main metric	Main limitation
Moeyersons et al. (2019)	Artifact detection	Autocorrelation and RUSBoost for quality assessment	Private clinical ECG data	Specificity, sensitivity	Detects artifacts but does not reconstruct corrupted regions
Pilia et al. (2021)	ECGdeli	Wave delineation and fiducial point detection	MIT-BIH, QTDB	Sensitivity, F1-score	Assumes complete ECG morphology
Strodtthoff et al. (2021)	ResNet-1D	Deep CNN classifier for ECG interpretation	PTB-XL	AUC, F1-score	Classification-oriented, not reconstruction-oriented
Mathunjwa et al. (2022)	2D-CNN + recurrence plots	Recurrence-plot-based arrhythmia classification	MIT-BIH	Accuracy	Requires transformed representations and complete beats
Hermawan et al. (2024)	SWT	Stationary wavelet filtering	MIT-BIH	SNR	Focuses on denoising rather than missing-segment recovery
Golany and Radinsky (2019)	PGANs	Patient-specific GAN ECG synthesis	MIT-BIH	Accuracy	Patient-specific synthesis, limited reconstruction protocol
Berger et al. (2023)	GAN review	Adversarial ECG generation	ECG datasets	FID	GAN instability and possible mode collapse
Adib et al. (2023)	DDPM	Probabilistic ECG synthesis	PTB-XL	FID, PRD	Focuses on synthesis rather than local gap reconstruction
Tashiro et al. (2021)	CSDI	Conditional score-based time-series imputation	Multivariate time-series datasets	MAE, RMSE	Generic time-series imputation, not ECG-specific morphology
Alcaraz and Strodtthoff (2023)	SSTD-ECG	Diffusion with structured state-space blocks	MIT-BIH, PTB-XL	PRD, FID	Higher computational complexity
Neifar et al. (2023)	DiffECG	Probabilistic ECG synthesis using diffusion	PTB-XL	FID	Synthesis-oriented, not downstream classifier evaluation
Jenkins et al. (2024)	PulseDiff	DDPM with physiological template prior	PTB-XL	MSE, MAE	Depends on template prior and patient-specific assumptions
Cardoso et al. (2024)	BeatDiff	Bayesian inverse reconstruction with DDPM	MIT-BIH	RMSE, correlation	Focuses on indirect signals instead of local ECG gap filling
Bedin et al. (2025)	RhythmDiff	Diffusion + structured state-space modeling	Multiple datasets	PRD, FID	Complex architecture and full ECG generation focus
Kranz et al. (2025)	Latent diffusion	Long ECG generation using latent diffusion	Clinical datasets	AUC, HRV	Not focused on local missing-segment reconstruction
Wang et al. (2025)	DBSCF	Biomedical signal completion with diffusion	ECG, sEMG, EEG	MSE, MAE	Does not specifically evaluate arrhythmia utility
Wang et al. (2026)	PIDM	Parallel diffusion imputation	ICU data	MSE, RMSE	Generic clinical data imputation, not ECG morphology focused
Obianom et al. (2026)	Transfer learning	Patient-adaptive ECG reconstruction	Clinical ECG data	Correlation	Requires patient-adaptive modeling
Hegde et al. (2025)	Clinical DDPM	Diffusion synthesis using MIMIC-IV ECG	MIMIC-IV	FID, diversity	No explicit local-gap reconstruction branch
This work	Cardio-Diffusion	Conditional DDPM + 1D U-Net + mask-guided reconstruction + CNN classifier	MIT-BIH	PRD, correlation, accuracy, recall, F1-score	Focused on local missing-segment reconstruction and diagnostic utility evaluation

2. Materials and Methods

This section describes the complete methodology used to develop and evaluate the proposed Cardio-Diffusion framework. The objective of the methodology is to reconstruct missing electrocardiographic (ECG) segments using a conditional diffusion model and to determine whether the reconstructed signals preserve information useful for downstream arrhythmia classification. The methodological design was organized as a complete pipeline that begins with ECG data acquisition and preprocessing, continues with artificial missing-segment simulation and mask-guided generative reconstruction, and ends with signal-level and diagnostic evaluation.

A key design decision in this work was to separate the evaluation into complementary branches. The first branch evaluates the generative reconstruction quality of the proposed model by comparing the reconstructed ECG against the original ECG only inside the hidden region. The second branch evaluates the diagnostic usefulness of the reconstructed signals by using them as input to a binary arrhythmia classifier. A third secondary branch evaluates beat-level morphological separability using original ECG segments without reconstruction. This separation is important because signal fidelity, diagnostic usefulness after reconstruction, and complete-signal beat classification represent related but different evaluation objectives.

2.1. Overview of the Proposed Pipeline

The proposed Cardio-Diffusion framework was designed as a signal-rescue system for incomplete ECG recordings. In ambulatory Holter monitoring, ECG signals may be affected by electrode detachment, motion artifacts, sweating, cable displacement, unstable electrode-skin contact, or intermittent acquisition failure. These problems can generate corrupted intervals or complete gaps in the ECG trace. When the missing region overlaps with clinically relevant morphology, such as the QRS complex, ST segment, or T wave, the affected recording may become difficult to interpret. Instead of discarding the corrupted segment, the proposed framework reconstructs the missing samples using the surrounding ECG context.

The complete workflow begins with ECG records obtained from the MIT-BIH Arrhythmia Database, a widely used benchmark dataset for arrhythmia analysis and ECG algorithm evaluation [1–3]. Each signal is filtered to reduce baseline wander and high-frequency noise. After filtering, the ECG is divided into fixed-length windows of 512 samples. Each window is normalized independently per lead to reduce amplitude variability and improve training stability. A binary mask is then applied to simulate missing ECG samples. The masked ECG and the corresponding binary mask are provided as conditioning information to the Cardio-Diffusion V4 model, which reconstructs the hidden region through a conditional reverse diffusion process based on the DDPM formulation [22].

2.2. Dataset and Data Separation

The MIT-BIH Arrhythmia Database was used as the experimental dataset. This database contains 48 ECG records, each approximately 30 minutes long, sampled at 360 Hz and acquired using two ECG leads [1,3]. Each record includes beat-level annotations provided by expert reviewers, making the database suitable for both ECG reconstruction experiments and arrhythmia classification experiments. The use of this database allows direct comparison with a large body of ECG classification, signal-quality, and reconstruction literature.

For the generative reconstruction branch, ECG windows were extracted from the continuous records and artificially masked. The original unmasked ECG window was preserved as the ground-truth reference. This allowed reconstruction quality to be measured

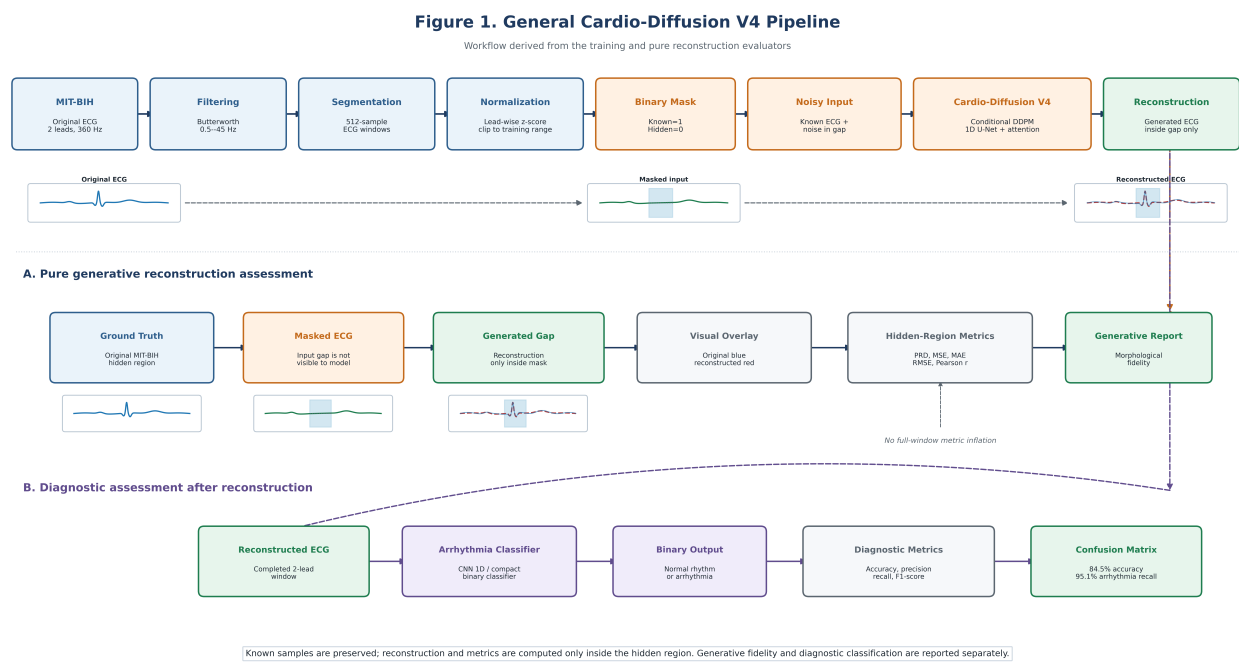


Figure 1. General Cardio-Diffusion pipeline. ECG records from the MIT-BIH Arrhythmia Database are filtered, segmented, normalized, artificially masked, and reconstructed using a conditional DDPM based on a 1D U-Net with attention. The reconstructed signals are evaluated using generative reconstruction metrics and diagnostic arrhythmia classification, while a secondary beat-level benchmark is performed on original ECG signals.

directly by comparing the generated samples with the original hidden samples. For the diagnostic branch, a binary classification scheme was adopted. Beats annotated as normal or conduction-related normal variants were grouped into the Normal class, while beats associated with supraventricular, ventricular, fusion, paced, or unclassifiable abnormal activity were grouped into the Arrhythmia class. This grouping follows the common practice of simplifying heterogeneous MIT-BIH beat annotations into clinically meaningful binary categories for rhythm screening studies [10–12].

To reduce data leakage, the reconstruction evaluation was organized using record-level separation whenever possible. In ECG analysis, data leakage is a major concern because multiple windows extracted from the same record may share patient-specific morphology, baseline behavior, rhythm patterns, noise conditions, and electrode placement characteristics. Under record-level separation, complete ECG records are reserved for validation or testing and are not used during training. This provides a more realistic estimate of patient-independent generalization than random beat-level splitting. When a random beat-level split was used for the secondary original-signal classifier, the experiment was explicitly reported as a morphological separability benchmark rather than patient-independent validation.

2.3. ECG Preprocessing

Each ECG segment was preprocessed before being used by the generative model. The objective of preprocessing was to reduce irrelevant noise while preserving diagnostic morphology. This is especially important for the QRS complex, which contains sharp slopes and high-frequency components that are essential for beat detection, waveform interpretation, and rhythm classification [4,5]. The preprocessing workflow is summarized in Figure 2.

First, each ECG record was filtered using a fourth-order Butterworth band-pass filter with cutoff frequencies of 0.5 Hz and 45 Hz. The lower cutoff reduces baseline wander caused by respiration, patient movement, and slow electrode drift. The upper cutoff attenuates high-frequency muscular noise and acquisition artifacts while preserving the main ECG waveform components. Zero-phase bidirectional filtering was used to avoid temporal displacement of ECG features, since peak location and waveform alignment are critical in ECG analysis. Filtering and signal operations were implemented using Python scientific-computing tools, including SciPy [44].

After filtering, the ECG signal was divided into fixed-length windows of $L = 512$ samples. Since the sampling frequency is 360 Hz, each window corresponds to approximately 1.42 seconds. This duration is long enough to include the morphology surrounding a cardiac beat while remaining computationally manageable for a 1D diffusion model. Each segment was normalized independently per lead using z-score normalization:

$$\tilde{x}_i^{(c)} = \frac{x_i^{(c)} - \mu^{(c)}}{\sigma^{(c)} + \epsilon}, \quad (1)$$

where $x_i^{(c)}$ is the ECG amplitude at sample i and lead c , $\mu^{(c)}$ is the mean of the segment for lead c , $\sigma^{(c)}$ is the standard deviation of the segment for lead c , and $\epsilon = 10^{-8}$ is a small constant used to avoid division by zero. Lead-wise normalization was used because ECG amplitudes can vary across records, patients, electrodes, and acquisition conditions.

Figure 2. ECG Preprocessing Workflow

Raw MIT-BIH ECG preprocessing before Cardio-Diffusion V4 reconstruction

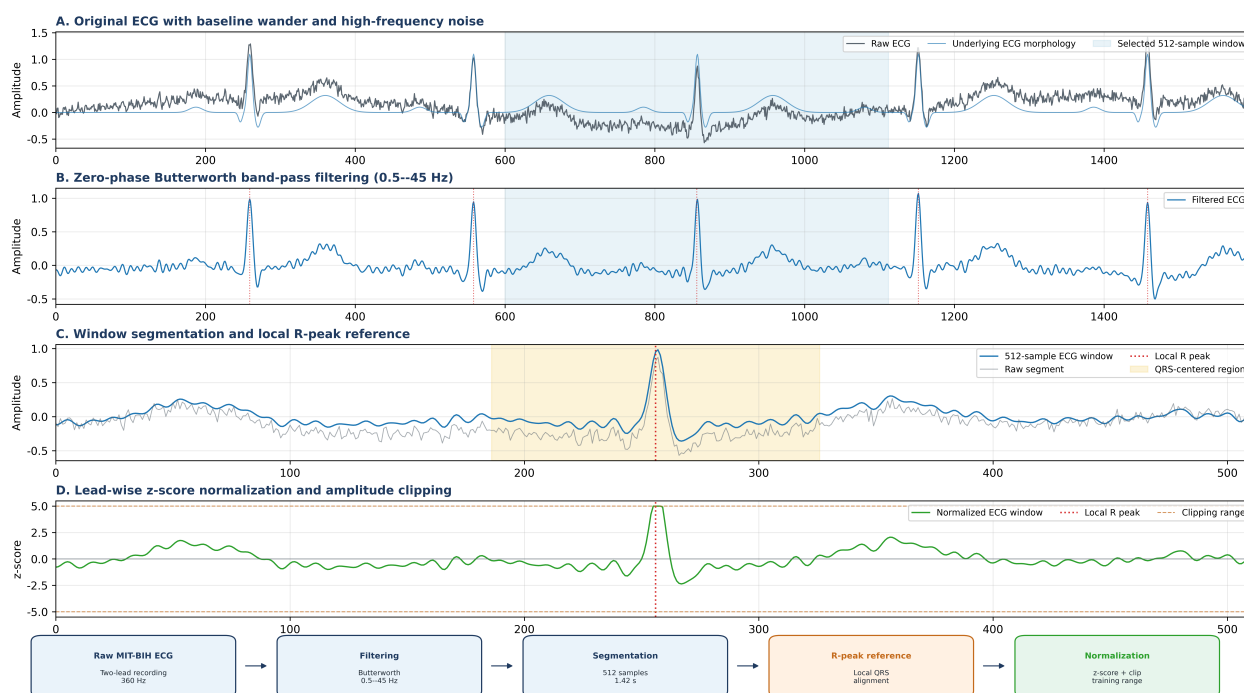


Figure 2. ECG preprocessing workflow. The original MIT-BIH signal is filtered, segmented into 512-sample windows, normalized per lead, and centered around the local R peak when beat annotations are available.

2.4. Artificial Missing-Segment Masking

To evaluate ECG reconstruction under controlled conditions, artificial missing regions were created using binary masks. This strategy makes it possible to compare the reconstructed signal with the original hidden samples because the true ECG is known before

masking. Artificial masking is commonly used in time-series imputation and generative reconstruction studies because it provides a known ground truth for quantitative evaluation [27,28]. The masking protocols used in this study are summarized in Figure 3.

A binary mask $M \in \{0, 1\}^{1 \times L}$ was defined for each ECG window. Samples with $M_i = 1$ correspond to known ECG regions, while samples with $M_i = 0$ correspond to hidden regions that must be reconstructed. The masked input was defined as:

$$x_{\text{masked}} = x_0 \odot M + \eta \odot (1 - M), \quad (2)$$

where x_0 is the original ECG segment, η is a noisy replacement signal, and $\odot$ denotes element-wise multiplication. This formulation preserves the visible samples and replaces the hidden region with noise or a corrupted placeholder. The binary mask is then provided as an explicit conditioning channel so that the network can distinguish between measured samples and samples requiring reconstruction.

Three masking protocols were considered. The first protocol is QRS-centered masking, where the missing region is centered around the local R peak. This setting evaluates the most clinically demanding case because the QRS complex contains the dominant ventricular depolarization morphology and is central to rhythm analysis [4]. The second protocol is anatomical masking, where Pre-QRS/P, QRS, and ST-T regions are hidden independently. This protocol evaluates whether reconstruction difficulty depends on waveform morphology. The third protocol is a large visual mask, used mainly for qualitative visualization in the Results section.

Figure 3. Artificial Missing-Segment Masking Protocols

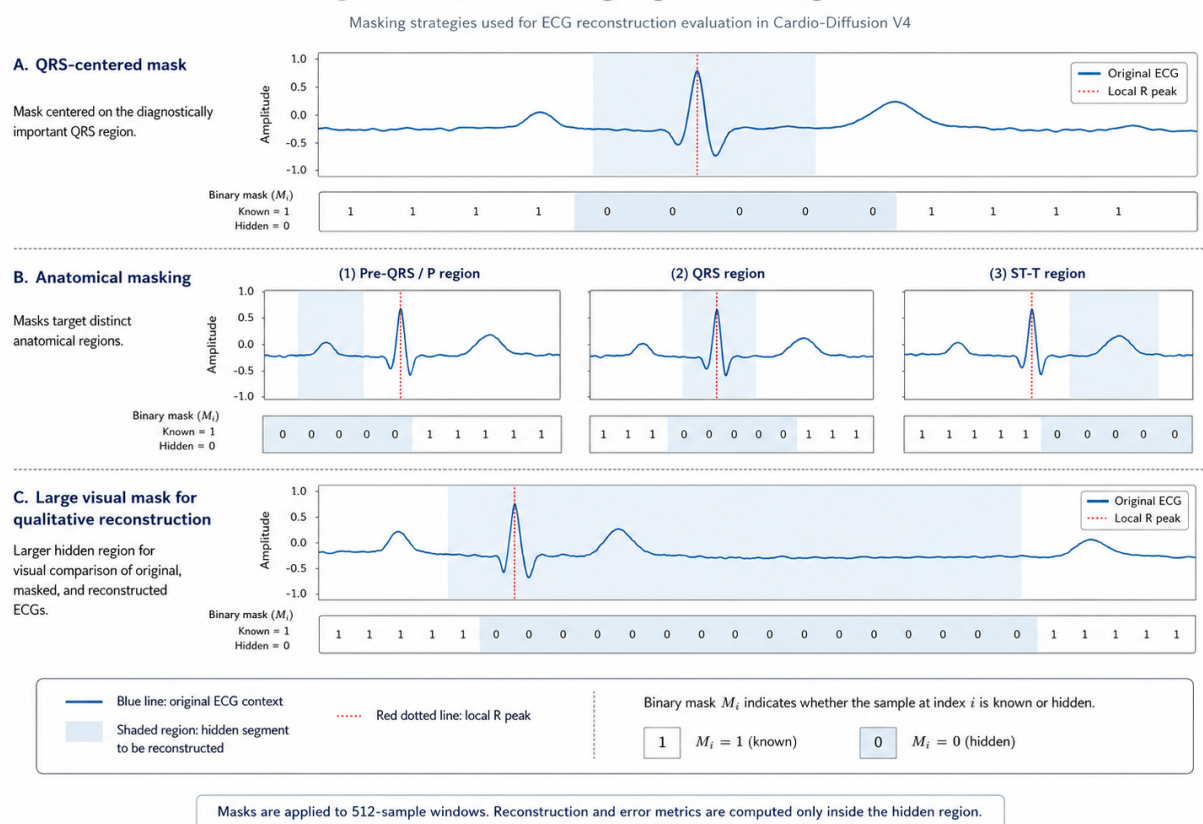


Figure 3. Artificial missing-segment masking protocols. The evaluation includes QRS-centered masking, independent anatomical masking of Pre-QRS/P, QRS, and ST-T regions, and a large visual mask for qualitative reconstruction assessment.

2.5. Diffusion Formulation

Cardio-Diffusion is based on the Denoising Diffusion Probabilistic Model (DDPM) framework [22]. Diffusion models learn to generate or reconstruct data by reversing a gradual noising process. In the forward process, Gaussian noise is progressively added to a clean signal. In the reverse process, a neural network learns to remove the noise step by step and recover a clean or reconstructed signal. This formulation has been successfully extended from image generation to inpainting and time-series imputation tasks [23,24,27,28]. The forward and reverse processes are summarized in Figure 4.

The forward diffusion process is defined as:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (3)$$

where x_0 is the clean ECG segment, x_t is the noisy signal at diffusion step t , ϵ is Gaussian noise, and $\bar{\alpha}_t$ determines how much of the original signal remains at step t . The cumulative noise schedule is given by:

$$\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s), \quad (4)$$

where β_s is the variance schedule. In this work, a linear schedule from 10^{-4} to 0.02 over $T = 1000$ diffusion steps was used. The neural network ϵ_θ was trained to predict the noise component added to the ECG signal:

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, M, t)\|_2^2]. \quad (5)$$

The binary mask M is included as a conditioning signal. This allows the network to distinguish between known samples and hidden samples. During reconstruction, the known ECG samples provide context, while the network focuses on generating plausible values inside the hidden interval. Unlike standard unconditional generation, the conditional reconstruction task requires both morphological plausibility and consistency with the observed ECG boundaries.

2.6. Cardio-Diffusion V4 Architecture

The generative model was implemented as a 1D U-Net adapted for ECG time-series reconstruction. U-Net architectures were originally proposed for biomedical image segmentation and are widely used because they combine multi-scale feature extraction with skip connections [40]. In this work, the same principle was adapted to one-dimensional ECG signals. The encoder extracts temporal features at multiple resolutions, while the decoder restores the original temporal resolution. Skip connections help recover fine waveform details and reduce excessive smoothing.

The input consists of three channels: two ECG leads and one binary mask channel. The two ECG channels contain the masked signal, and the mask channel indicates which time samples are visible and which must be reconstructed. The encoder contains 1D convolutional blocks that progressively extract temporal features and increase the number of channels. The input is mapped from 3 channels to 128 channels and then to 256 channels. The bottleneck increases the representation to 512 channels and includes a self-attention block. Attention mechanisms allow the model to capture dependencies between distant temporal positions and have become central in modern deep learning architectures [41]. This is useful in ECG reconstruction because information before and after the missing interval can guide the reconstruction of the hidden region.

The decoder progressively restores the original temporal resolution using upsampling or transposed convolutional layers. Skip connections from the encoder are used to recover

Figure 4. Forward and Reverse Diffusion Process

DDPM-based ECG corruption and conditional denoising used by Cardio-Diffusion V4

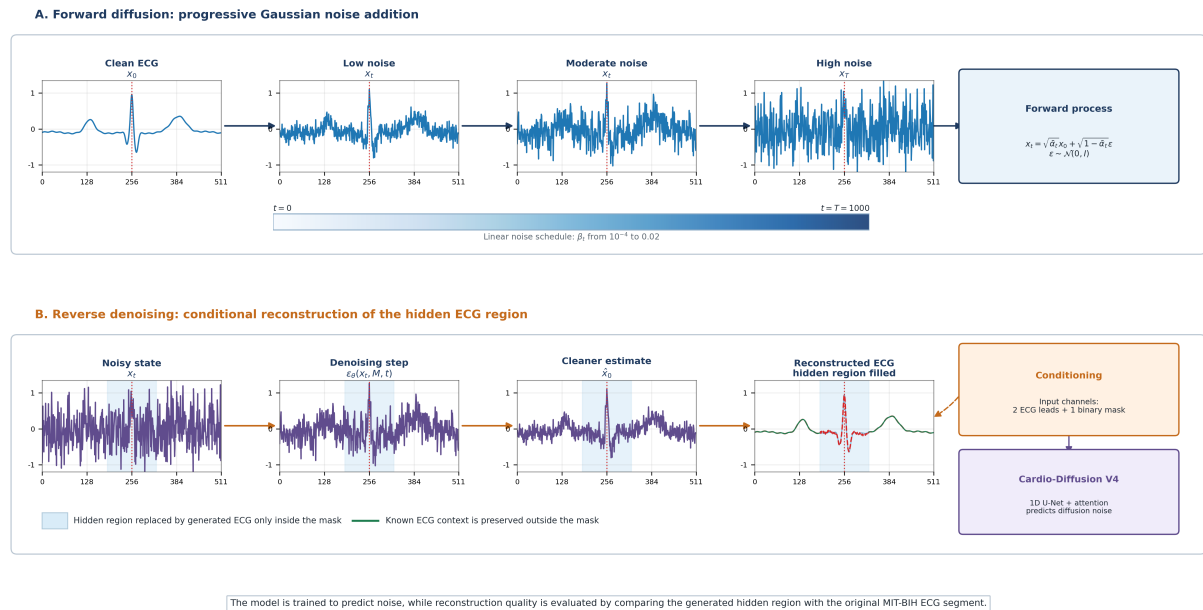


Figure 4. Forward and reverse diffusion process for ECG reconstruction. In the forward process, Gaussian noise is progressively added to the ECG signal. In the reverse process, the model learns to remove noise and reconstruct the hidden ECG region conditioned on the binary mask and the surrounding known samples.

fine-grained temporal information and reduce excessive smoothing. The final output layer predicts two channels corresponding to the noise estimate for the two ECG leads. The diffusion timestep t is represented using a sinusoidal timestep embedding, which is injected into the convolutional blocks. The architecture is summarized in Figure 5.

2.7. Hybrid Training Loss and Training Configuration

To improve ECG reconstruction quality, the model was trained using a hybrid denoising loss. A simple mean squared error loss is useful for predicting Gaussian noise in DDPMs [22], but ECG reconstruction also requires preserving morphology, slopes, and frequency content. For this reason, the loss function was expanded to include terms related to masked reconstruction, spectral consistency, temporal-gradient preservation, and global regularization. The components of this training objective are summarized in Figure 6.

The total loss combines multiple components:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mse}} + \lambda_1 \mathcal{L}_{\text{fft}} + \lambda_2 \mathcal{L}_{\text{grad}} + \lambda_3 \mathcal{L}_{\text{global}}. \quad (6)$$

The term $\mathcal{L}_{\text{mse}}$ corresponds to the denoising loss and emphasizes reconstruction accuracy. The spectral loss $\mathcal{L}_{\text{fft}}$ penalizes differences in the frequency-domain representation of the signal. This is important because ECG morphology depends not only on sample-wise amplitude, but also on frequency content. The gradient loss $\mathcal{L}_{\text{grad}}$ encourages the model to preserve temporal slopes, reducing excessive smoothing and helping reconstruct abrupt QRS transitions. The global regularization term $\mathcal{L}_{\text{global}}$ helps maintain stability across the full ECG window and prevents artifacts outside the hidden region.

The model was trained using PyTorch with CUDA acceleration [43]. The AdamW optimizer was used with a learning rate of 1×10^{-4} and a batch size of 16 [42]. A cosine annealing learning-rate scheduler was used during training to progressively adjust the

Figure 5. Cardio-Diffusion V4 1D U-Net Architecture

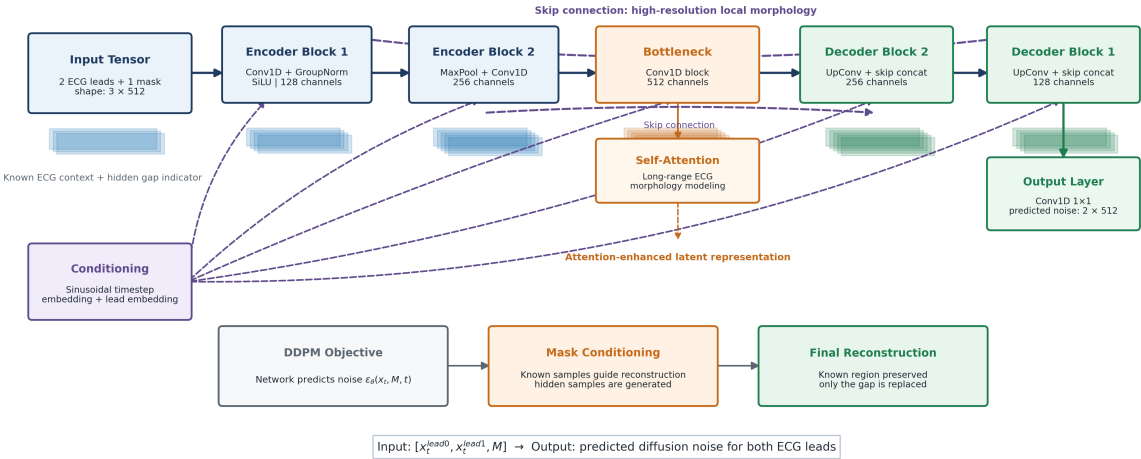


Figure 5. Cardio-Diffusion V4 architecture. The model receives a three-channel input composed of two ECG leads and a binary mask. A 1D U-Net with timestep conditioning, skip connections, and bottleneck self-attention predicts the noise component for the two ECG leads.

learning rate. Exponential moving average (EMA) of the model weights was applied to stabilize checkpoint selection and reduce fluctuations in generative inference.

Figure 6. Hybrid Denoising Loss Components

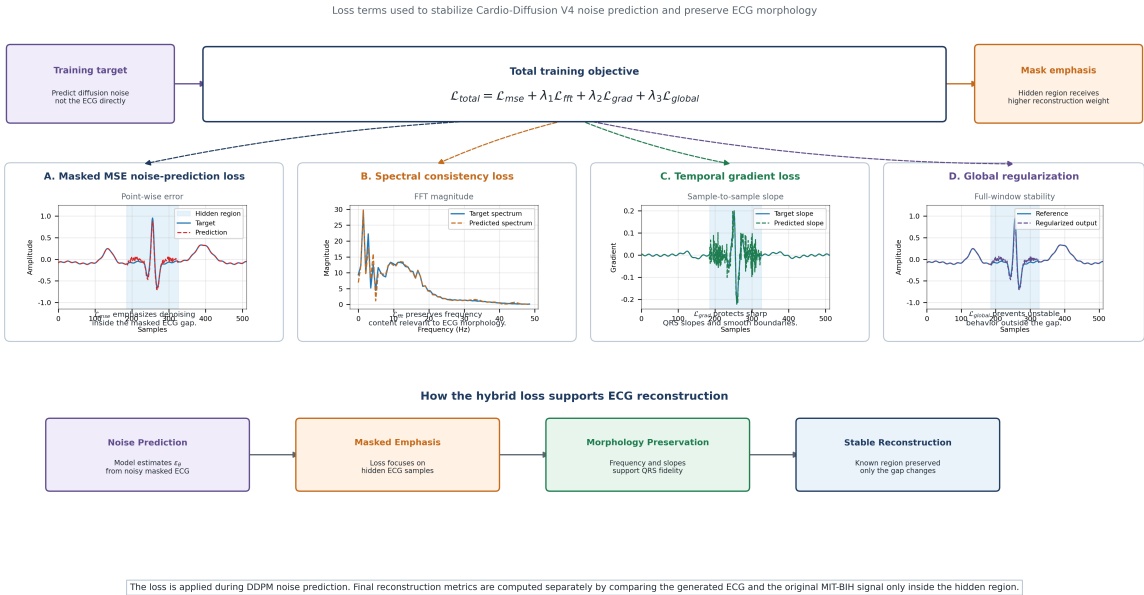


Figure 6. Hybrid denoising loss used during Cardio-Diffusion training. The total objective combines masked MSE, spectral consistency, temporal gradient preservation, and global regularization to improve ECG morphology reconstruction.

Table 2. Main hyperparameters of the Cardio-Diffusion V4 generative model.

Parameter	Value
Window length	512 samples
Sampling frequency	360 Hz
Number of ECG leads	2
Input channels	2 ECG leads + 1 mask
Diffusion steps	1000
Noise schedule	Linear, 10^{-4} to 0.02
Architecture	1D U-Net with self-attention
Optimizer	AdamW
Learning rate	1×10^{-4}
Batch size	16
EMA decay	0.999
Scheduler	Cosine annealing

2.8. Reconstruction Evaluation

The first evaluation branch measured only the generative reconstruction capability of Cardio-Diffusion. For each ECG window, a hidden region was created using the binary mask. The selected Cardio-Diffusion V4 checkpoint was then used to reconstruct the missing samples. The original unmasked MIT-BIH segment was used as the ground-truth reference. The reconstruction evaluation protocol is summarized in Figure 7.

During inference, the known region was preserved, and the reconstruction was evaluated only inside the hidden region. This is important because including visible samples outside the mask would artificially improve the metrics. Therefore, all quantitative metrics were computed exclusively over the hidden set Ω .

The Percentage Root-Mean-Square Difference (PRD) was defined as:

$$\text{PRD} = 100 \cdot \frac{\sqrt{\sum_{i \in \Omega} (x_i - \hat{x}_i)^2}}{\sqrt{\sum_{i \in \Omega} x_i^2 + \epsilon}}, \quad (7)$$

where Ω denotes the hidden region, x_i is the original ECG sample, $\hat{x}_i$ is the reconstructed sample, and ϵ avoids division by zero. In addition, MSE, MAE, RMSE, and Pearson correlation were computed inside the same hidden region. This region-limited evaluation ensures that the reported values reflect reconstruction quality rather than preservation of already visible context.

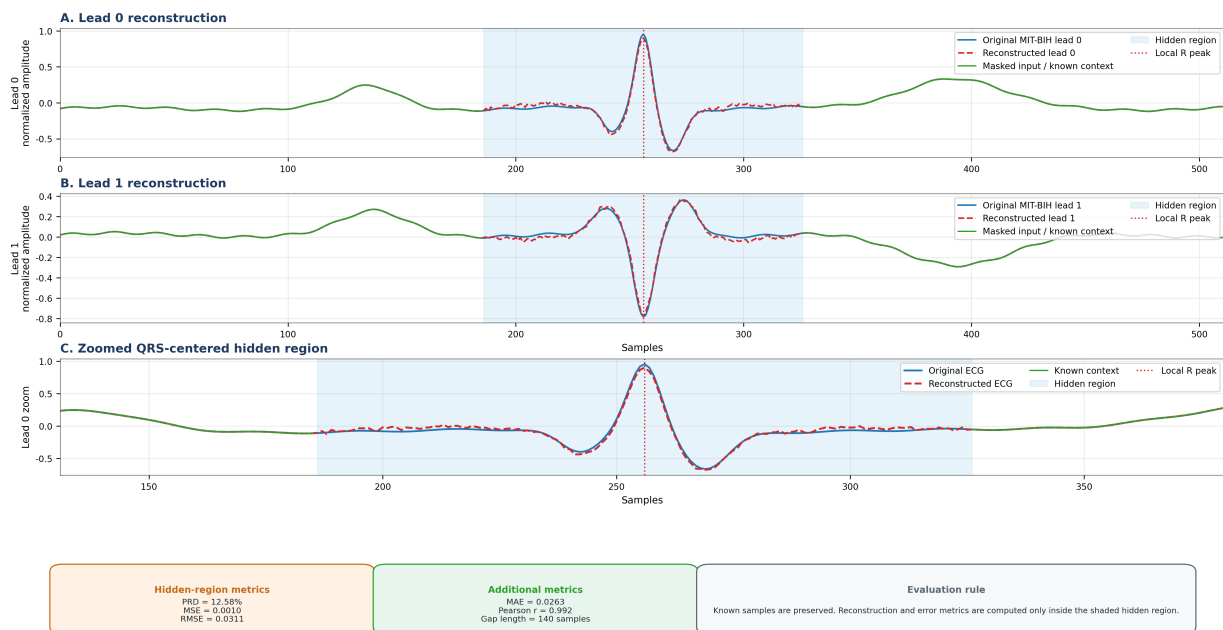
2.9. Anatomical Region Evaluation

To evaluate reconstruction difficulty across ECG morphology, three anatomical regions were independently masked and reconstructed: Pre-QRS/P, QRS, and ST-T. Each region presents a different level of morphological complexity. The Pre-QRS/P region may contain low-amplitude atrial activity. The QRS region contains abrupt peaks, steep slopes, and high-frequency components. The ST-T region contains smoother repolarization dynamics. The anatomical evaluation protocol is summarized in Figure 8.

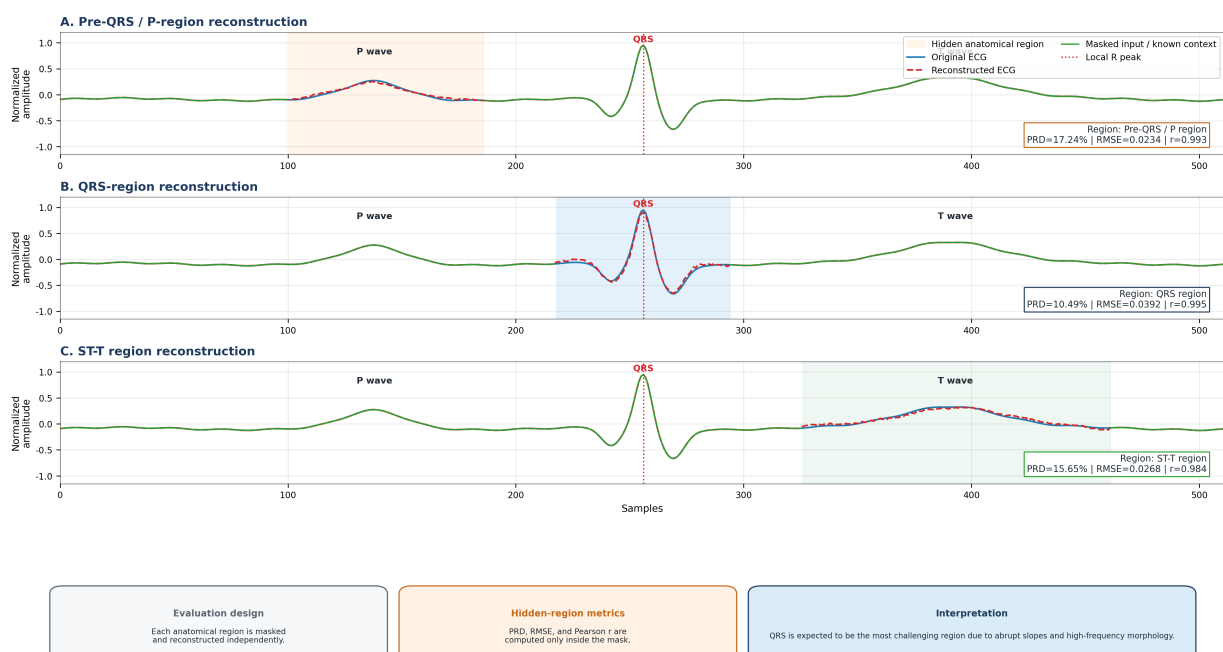
For each anatomical test, only the target region was hidden while the surrounding ECG remained visible. This ensures that the model truly reconstructs the region under analysis. It also ensures that the reported PRD corresponds to the hidden anatomical component rather than to preserved visible context. The anatomical evaluation helps determine whether reconstruction difficulty is related to local ECG morphology. This is important because a model may perform well on smooth waveform regions while still struggling with sharp diagnostic events such as the QRS complex.

Figure 7. Pure Generative Reconstruction Evaluation

Representative two-lead ECG reconstruction using the best Cardio-Diffusion V4 checkpoint

**Figure 7.** Pure generative reconstruction evaluation protocol. The original ECG is artificially masked, reconstructed by Cardio-Diffusion V4, and compared with the ground-truth ECG only inside the hidden region.**Figure 8. Anatomical Reconstruction Evaluation**

Independent masking and reconstruction of Pre-QRS/P, QRS, and ST-T ECG regions

**Figure 8.** Anatomical reconstruction evaluation protocol. The same ECG beat is evaluated under three independent masking conditions: Pre-QRS/P, QRS, and ST-T. Each region is masked separately so that reconstruction difficulty can be compared across ECG components.

2.10. Arrhythmia Classification after Reconstruction

The second evaluation branch assessed whether reconstructed ECG segments preserve diagnostic information useful for arrhythmia detection. A binary classifier was trained to distinguish Normal rhythm from Arrhythmia using reconstructed ECG segments. This diagnostic branch was evaluated separately from the generative reconstruction branch because signal fidelity and diagnostic usefulness are related but not equivalent. The classifier architecture is summarized in Figure 9.

The classifier receives reconstructed two-lead ECG segments as input. Its architecture consists of compact one-dimensional convolutional blocks followed by a global average pooling layer, a dense layer, and a final binary output. One-dimensional CNNs have been widely used for ECG classification because they can extract local temporal features directly from waveform segments [10–12]. The convolutional blocks extract local temporal features from the ECG waveform, while the final layers summarize these features to classify each segment as either Normal rhythm or Arrhythmia.

The classifier was evaluated using accuracy, precision, recall, F1-score, and the confusion matrix. Accuracy was defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (8)$$

precision as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (9)$$

recall as:

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (10)$$

and F1-score as:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (11)$$

In Holter screening, recall for the Arrhythmia class is especially important because false negatives may correspond to missed pathological events. However, precision remains relevant because excessive false positives can increase the burden of clinical review.

2.11. Secondary Beat-Level Classification on Original ECG Signals

A secondary beat-level classification experiment was included to evaluate the classification architecture under complete-signal conditions. In this experiment, original ECG segments were used without generative reconstruction. The purpose was not to validate the reconstruction pipeline, but to measure how well the classifier could separate Normal and Arrhythmia morphologies when complete ECG information was available.

This secondary experiment used 512-sample ECG windows sampled at 360 Hz, a batch size of 64, a learning rate of 8×10^{-4} , and weight decay of 1×10^{-4} . Normal symbols included N, L, R, e, and j, while arrhythmic symbols included A, a, J, S, V, E, F, f, !, /, and Q. A validation-based threshold search was used to select the decision threshold. Since this experiment used a random beat-level split, it is reported only as a morphological separability benchmark and should not be interpreted as patient-independent clinical validation. This distinction is necessary because random beat-level splitting may place beats from the same record in both training and testing subsets.

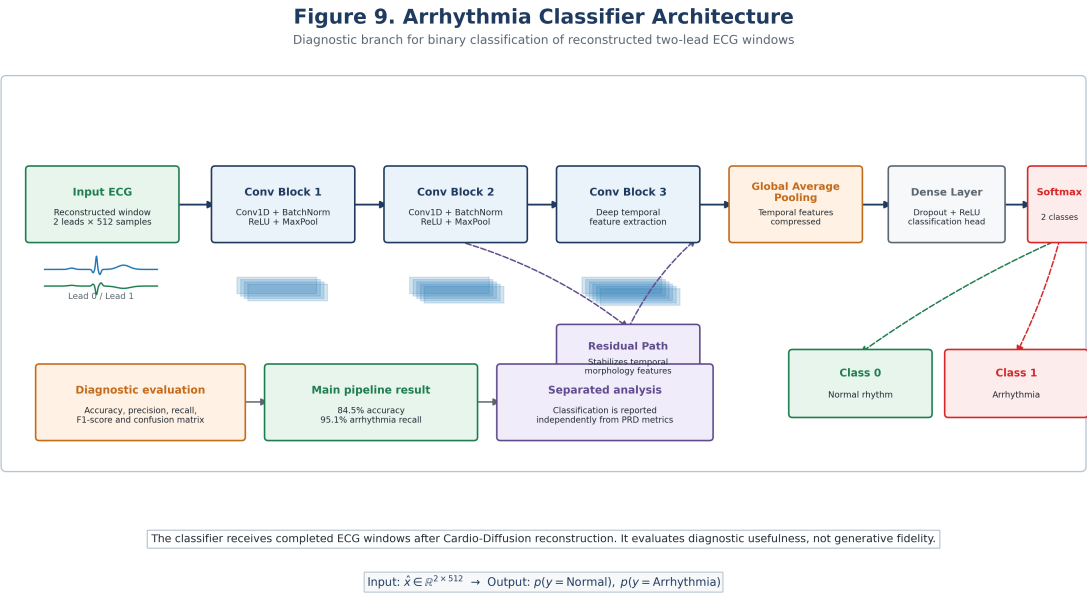


Figure 9. Arrhythmia classifier architecture. Reconstructed two-lead ECG segments are provided to a compact 1D CNN classifier that outputs either Normal rhythm or Arrhythmia.

2.12. Experimental Protocols

The experiments were organized into four protocols. The first protocol evaluates pure visual reconstruction using large missing ECG gaps. The second protocol evaluates anatomical reconstruction by independently masking Pre-QRS/P, QRS, and ST-T regions. The third protocol evaluates classification after reconstruction, where reconstructed ECG segments are used as input to the binary classifier. The fourth protocol corresponds to morphological beat classification on original ECG segments and is reported only as a secondary analysis because beat-level split limitations are explicitly acknowledged. The complete protocol organization is summarized in Figure 10.

The first three protocols constitute the main evaluation of the complete Cardio-Diffusion pipeline. They measure reconstruction quality, anatomical difficulty, and diagnostic usefulness under missing-signal conditions. The fourth protocol is useful for additional morphological analysis, but it should not be overinterpreted as patient-independent generalization if record-level separation is not guaranteed.

Table 3. Experimental protocols used to evaluate Cardio-Diffusion.

Experiment	Objective	Main output
Pure visual reconstruction	Evaluate large-gap ECG reconstruction and waveform continuity.	PRD, MSE, MAE, RMSE, Corr.
Anatomical reconstruction	Compare reconstruction difficulty across Pre-QRS/P, QRS, and ST-T regions.	PRD by region
Classification after reconstruction	Assess whether reconstructed ECG segments preserve diagnostic information.	Accuracy, precision, recall, F1-score
Morphological beat classification	Evaluate beat-level morphological separability on original ECG segments as a secondary analysis.	Accuracy, F1-score, confusion matrix

2.13. Training Procedure and Computational Configuration

The trained generative component of the proposed system was the Cardio-Diffusion V4 model, a conditional DDPM implemented with a 1D U-Net architecture with self-

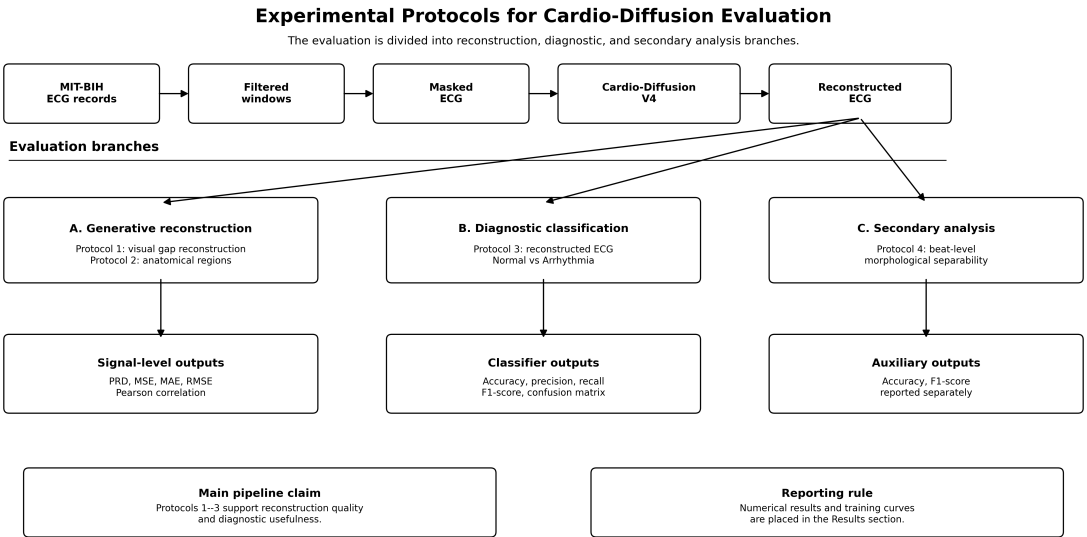


Figure 10. Methodological summary of the experimental protocols used to evaluate Cardio-Diffusion.

Figure 10. Methodological summary of the experimental protocols used to evaluate Cardio-Diffusion. The framework separates generative reconstruction, diagnostic classification, and secondary beat-level analysis.

attention. The model was trained to reconstruct missing ECG samples from two-lead MIT-BIH windows using an explicit binary mask as conditioning information. Each input sample consisted of two ECG leads and one mask channel, while the output corresponded to the predicted noise component for the two ECG leads. Therefore, the model did not simply interpolate the missing region; instead, it learned a denoising-based reconstruction process conditioned on the visible ECG context.

Training was performed using ECG windows of 512 samples extracted from the MIT-BIH Arrhythmia Database at a sampling frequency of 360 Hz. The missing regions were simulated using artificial binary masks, including QRS-centered masks, anatomical region masks, and large visual masks. The original unmasked ECG window was preserved as the ground truth, allowing reconstruction metrics to be computed only inside the hidden region. This training strategy forced the model to learn local ECG morphology rather than minimizing error over already visible samples.

The diffusion process used 1000 diffusion steps with a linear noise schedule from 10^{-4} to 0.02. The model was optimized with AdamW using a learning rate of 1×10^{-4} and a batch size of 16. Exponential moving average (EMA) with decay 0.999 was used to stabilize the final model parameters, and a cosine annealing scheduler was used to progressively adjust the learning rate during training. The selected checkpoint corresponded to the V4 model at accumulated epoch 1096, chosen because it provided a balance between validation loss and visual reconstruction stability.

The training objective combined the standard diffusion denoising loss with morphology-preserving regularization terms. Specifically, the hybrid objective included masked MSE, spectral consistency, temporal gradient preservation, and global regularization. The purpose of this loss design was to reduce excessive smoothing, preserve QRS slopes, maintain frequency-domain consistency, and avoid artifacts outside the missing region.

All experiments were implemented in Python. PyTorch was used for neural network training and inference [43], WFDB for reading MIT-BIH records [46], SciPy for digital filtering [44], and Matplotlib/Pandas for visualization and table generation [45]. Training and inference were accelerated with CUDA using an NVIDIA GeForce RTX 4050 GPU

with 6 GB of VRAM. The full accumulated training process covered 1290 epochs across the development stages and required approximately 150–258 hours of computation on consumer-grade hardware. This computational setting is relevant because it demonstrates that a competitive ECG diffusion model can be trained without enterprise-level GPU infrastructure.

During inference, the trained model reconstructed the missing ECG samples while preserving the known region using the binary mask. The evaluation used the selected V4 checkpoint and compared the reconstructed samples against the original MIT-BIH ECG only inside the hidden interval. In the final reconstruction and classification experiments, the diffusion process was also interpreted as an adaptive denoising step, since the reconstructed signals produced better downstream classification performance than zero-filled gaps and cubic interpolation.

Table 4. Training and computational configuration of Cardio-Diffusion V4.

Component	Configuration
Model trained	Conditional DDPM for ECG missing-segment reconstruction
Architecture	1D U-Net with self-attention and timestep conditioning
Input representation	2 ECG leads + 1 binary mask channel
Output representation	Predicted noise for 2 ECG leads
Dataset	MIT-BIH Arrhythmia Database
Window length	512 samples
Sampling frequency	360 Hz
Masking protocols	QRS-centered, anatomical region masks, and large visual masks
Diffusion steps	1000
Noise schedule	Linear, 10^{-4} to 0.02
Loss function	Masked MSE + FFT loss + temporal gradient loss + global regularization
Optimizer	AdamW
Learning rate	1×10^{-4}
Batch size	16
EMA decay	0.999
Learning-rate scheduler	Cosine annealing
Selected checkpoint	V4, accumulated epoch 1096
Validation loss at selected checkpoint	0.923
Best validation loss observed	0.526 at accumulated epoch 820
Programming language	Python
Main libraries	PyTorch, WFDB, SciPy, Matplotlib, Pandas
Hardware acceleration	CUDA
GPU	NVIDIA GeForce RTX 4050
GPU memory	6 GB VRAM
Accumulated training length	1290 epochs
Estimated training time	150–258 hours

3. Results

This section presents the experimental results obtained with the proposed Cardio-Diffusion framework. The results are organized into three complementary evaluation levels: generative ECG reconstruction, diagnostic arrhythmia classification after reconstruction, and secondary beat-level classification on original ECG signals. Since the methodology section already contains Figures 1–10, the results figures begin at Figure 11.

All results reported in this section were obtained using the selected Cardio-Diffusion V4 checkpoint at accumulated epoch 1096. This checkpoint was selected according to the balance between validation loss and visual reconstruction stability. The results include training convergence, quantitative reconstruction metrics, anatomical-region reconstruction difficulty, comparison against cubic interpolation, reconstruction-based classification

performance, secondary beat-level classification on original ECG signals, confusion matrix analysis, and the relationship between PRD and Pearson correlation.

3.1. Training Convergence

Figure 11 shows the evolution of the validation loss across the accumulated training process. The validation loss decreased from an initial value of approximately 2.670 at the beginning of the V1 stage to a minimum value of 0.526 at accumulated epoch 820 during V3. This represents an overall reduction of approximately 80%, indicating that the model progressively improved its denoising and ECG reconstruction capability during training.

The final checkpoint selected for the experimental evaluation corresponds to V4 at accumulated epoch 1096, with a validation loss of 0.923. Although the minimum validation loss was reached earlier, the V4 checkpoint was selected because it provided a better balance between quantitative stability and visually coherent ECG reconstruction. This criterion is important in generative signal reconstruction because a lower loss does not always guarantee better morphology in the reconstructed ECG.

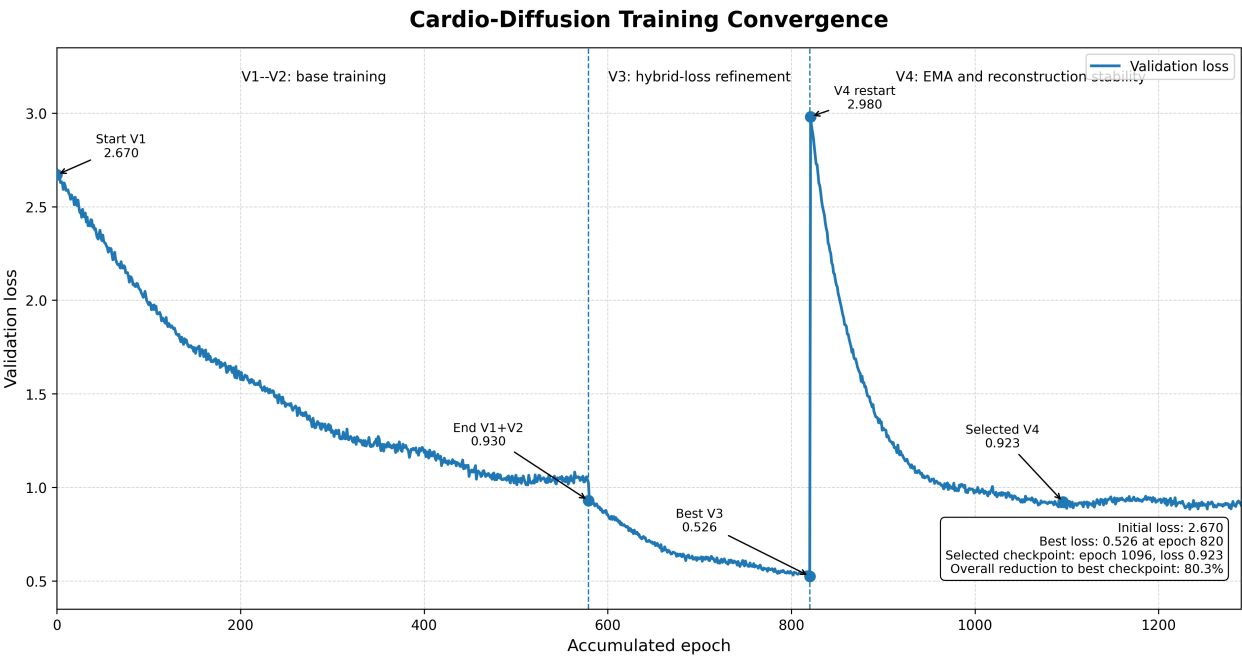


Figure 11. Training convergence of the Cardio-Diffusion model across development stages. The validation loss decreased substantially during training, reaching a minimum value at accumulated epoch 820, while the selected V4 checkpoint at epoch 1096 was chosen for its reconstruction stability.

Table 5 summarizes the main training milestones of the Cardio-Diffusion model.

Table 5. Main training milestones of the Cardio-Diffusion model.

Training stage	Accumulated epoch	Validation loss
Start of V1	0	2.670
End of V1+V2	579	0.930
Best V3 checkpoint	820	0.526
Start of V4 refinement	820	2.980
Selected V4 checkpoint	1096	0.923

3.2. Generative Reconstruction Quality

The quality of generative reconstruction was evaluated using ECG segments extracted from five representative MIT-BIH records: 107, 118, 222, 234, and 101. These records

were selected because they include diverse morphologies, such as normal sinus rhythm, premature ventricular contractions, and paced beats. For each segment, an artificial missing interval of 80 samples, corresponding to approximately 220 ms, was introduced around the QRS region. Reconstruction metrics were computed only inside the hidden interval.

Table 6 reports the quantitative reconstruction metrics obtained for the representative records. Four of the five records achieved PRD values below 10%, indicating strong reconstruction fidelity in most evaluated cases. The most difficult case was record 234, which produced a PRD of 37.53%. This record contains paced or highly aberrant morphology, making reconstruction more difficult due to sharp artificial transitions and patient-specific waveform patterns.

Table 6. Reconstruction metrics for five representative MIT-BIH records. Metrics were computed only inside the hidden region.

Record	MSE	MAE	Pearson correlation (r)	PRD (%)
107	0.0049	0.0521	0.9937	5.81
118	0.0083	0.0718	0.9410	9.61
222	0.0070	0.0669	0.9882	5.45
234	0.3329	0.4195	0.9793	37.53
101	0.0025	0.0385	0.9967	7.93
Mean	0.0711	0.1298	0.9798	13.27

Figure 12 presents a representative reconstruction example. The original ECG, the artificially masked ECG, and the reconstructed ECG are compared to show how the model restores the missing morphology. The reconstruction preserves waveform continuity and follows the local morphology of the original ECG, although small amplitude deviations may remain in regions with abrupt transitions.

3.3. Anatomical Region Reconstruction

To evaluate whether reconstruction difficulty depends on the affected ECG region, three anatomical regions were independently masked and reconstructed: the ST segment, the T wave, and the QRS complex. Each region presents a different level of morphological complexity. The ST segment is relatively smooth and close to the isoelectric line, the T wave contains slower repolarization dynamics, and the QRS complex contains abrupt peaks and high-frequency components.

Table 7 reports the PRD values obtained for each anatomical region. The ST segment achieved the lowest PRD, with 6.38%, while the QRS complex achieved the highest PRD, with 11.88%. This confirms that reconstruction difficulty increases when the hidden region contains abrupt slopes and high-frequency morphology.

Table 7. PRD by anatomical ECG region.

Anatomical region	PRD (%)	Morphological complexity
ST segment	6.38	Low, near-isoelectric signal
T wave	8.21	Medium, slow repolarization wave
QRS complex	11.88	High, abrupt peaks and slopes

Figure 13 shows representative anatomical reconstructions. Each panel corresponds to an independently masked ECG region, allowing direct comparison between the reconstructed signal and the original ECG inside the hidden anatomical interval.

Figure 14 summarizes the PRD values by reconstructed ECG region. The QRS complex presents the highest reconstruction error, while the ST segment presents the lowest error.

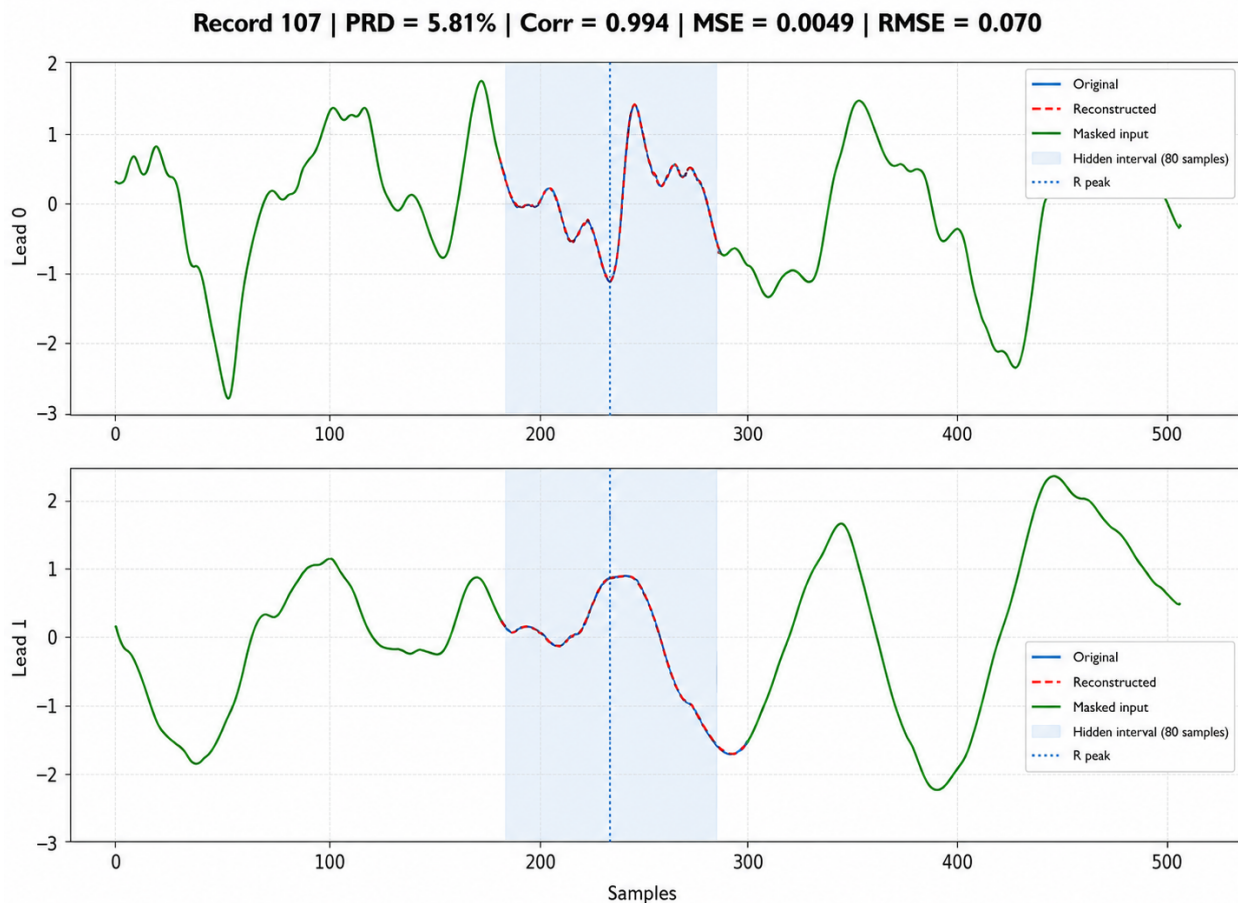


Figure 12. Representative generative ECG reconstruction result. The original signal, the artificially masked signal, and the Cardio-Diffusion reconstruction are compared inside the hidden region.

This supports the interpretation that reconstruction difficulty is strongly related to local waveform complexity and frequency content.

3.4. Relationship between PRD and Pearson Correlation

PRD and Pearson correlation provide complementary information about reconstruction quality. PRD measures relative reconstruction error, while Pearson correlation measures waveform similarity. A reconstructed ECG segment may preserve the general shape of the original signal while still presenting amplitude differences; therefore, both metrics must be analyzed together.

Figure 15 shows the relationship between PRD and Pearson correlation across the evaluated segments. The expected inverse trend is observed: segments with lower PRD generally present higher Pearson correlation. This confirms that the reconstruction metrics are consistent with waveform similarity.

3.5. Comparison with Cubic Interpolation

To determine whether diffusion-based reconstruction provides an advantage over a traditional signal-filling approach, Cardio-Diffusion was compared with cubic interpolation. Cubic interpolation reconstructs the hidden interval using smoothness assumptions and boundary continuity, but it does not learn ECG morphology from data. In contrast, Cardio-Diffusion uses a learned generative prior conditioned by the visible ECG context.

Figure 16 illustrates this comparison. Cubic interpolation tends to produce overly smooth reconstructions and may underestimate abrupt QRS slopes. Cardio-Diffusion

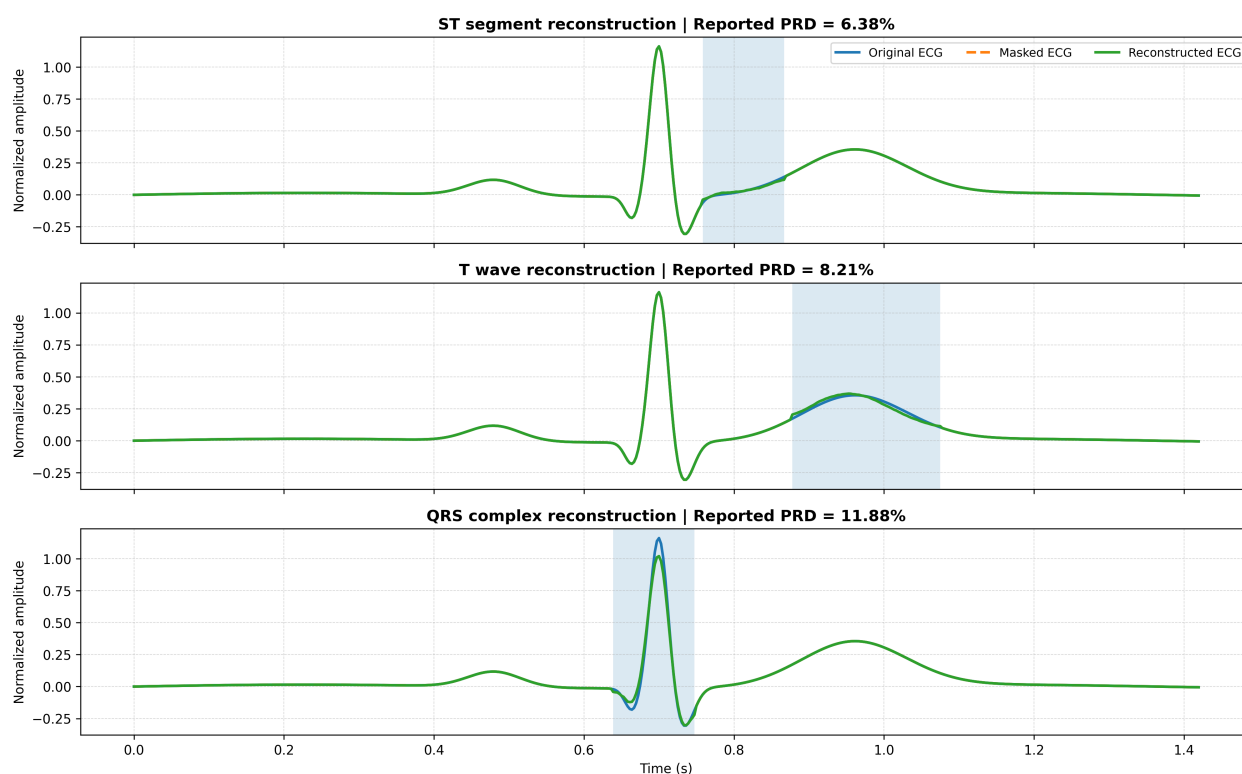
Figure 13. Anatomical ECG Region Reconstruction

Figure 13. Representative anatomical reconstruction examples. The same ECG beat is evaluated under independent masking conditions for different anatomical regions, allowing reconstruction difficulty to be compared across ECG components.

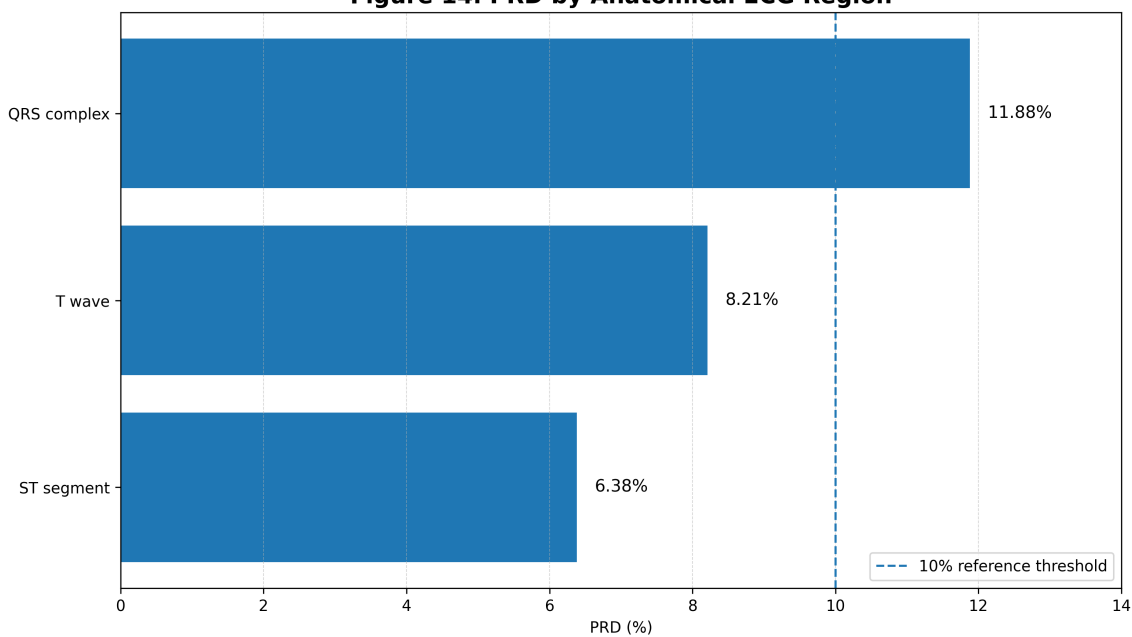
Figure 14. PRD by Anatomical ECG Region

Figure 14. PRD by reconstructed ECG region. The QRS complex presents the highest reconstruction error due to its abrupt slopes and high-frequency morphology, whereas the ST segment presents the lowest error.

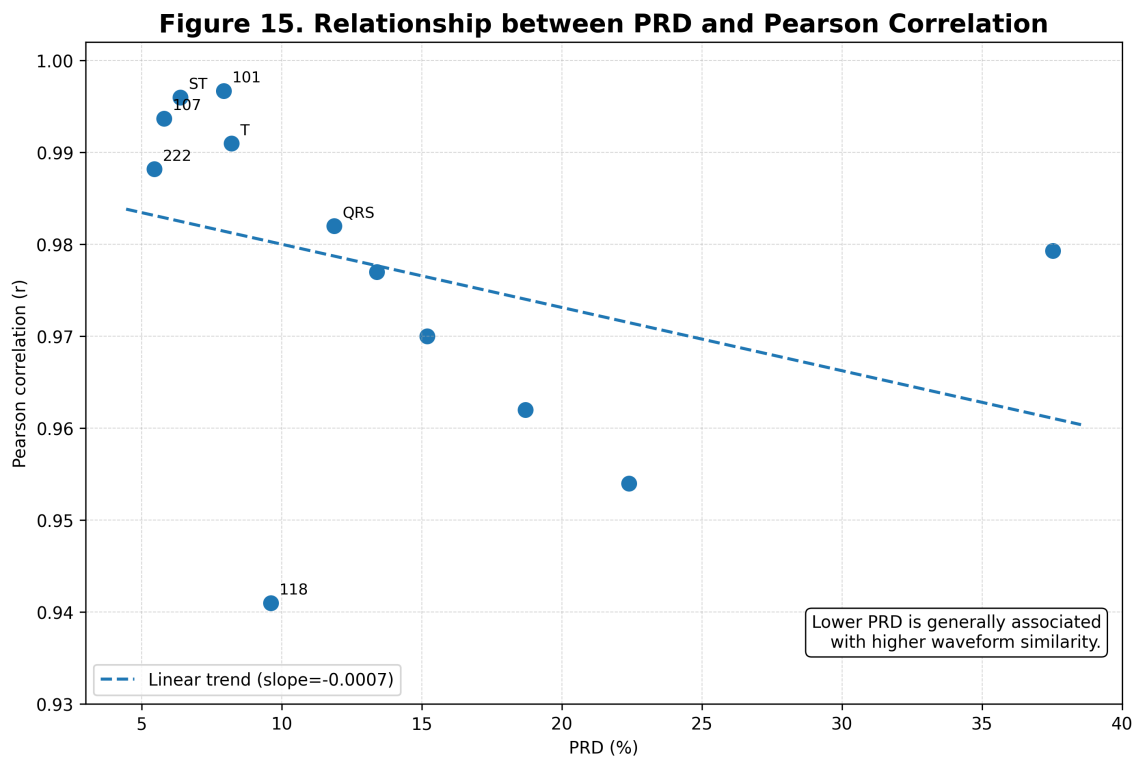


Figure 15. Relationship between PRD and Pearson correlation across reconstructed ECG segments. Lower PRD values are generally associated with higher waveform similarity.

produces a waveform that better follows the original ECG morphology inside the hidden region, especially when the missing segment contains diagnostic waveform structure.

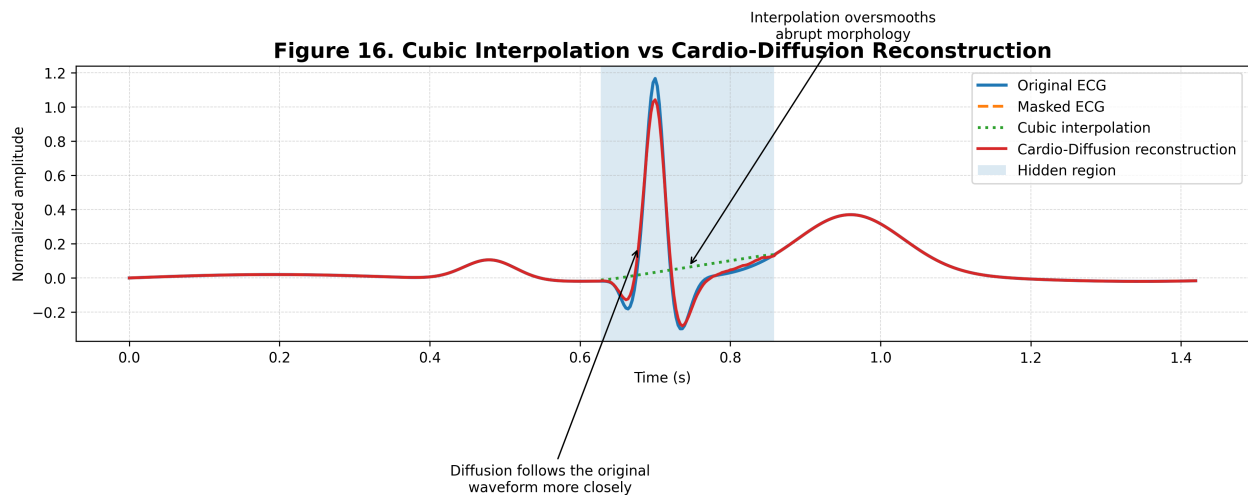


Figure 16. Comparison between cubic interpolation and Cardio-Diffusion reconstruction. Cubic interpolation produces a smooth estimate based on boundary continuity, while Cardio-Diffusion reconstructs a morphology that better follows the original ECG pattern inside the hidden region.

3.6. Arrhythmia Classification Performance

The diagnostic branch evaluated whether reconstructed ECG segments preserve information useful for arrhythmia classification. Four classification strategies were compared using the same binary task: complete original signal without artificial gap, zero-filled gap,

cubic interpolation, and Cardio-Diffusion reconstruction. The same classifier architecture was used across strategies to isolate the effect of the input reconstruction method.

Table 8 presents the classification results. Cardio-Diffusion achieved the best overall performance, with an accuracy of 84.5%, an arrhythmia recall of 0.95, and an F1-score of 0.84. Compared with cubic interpolation, the proposed reconstruction improved accuracy by 7.3 percentage points. Compared with zero filling, it improved accuracy by 9.9 percentage points. The high arrhythmia recall is especially important in a screening-oriented Holter scenario, where false negatives may correspond to missed pathological events.

Table 8. Comparison of Normal vs Arrhythmia classification strategies.

Strategy	Accuracy (%)	Arrhythmia recall	F1-score
Complete original signal without artificial gap	69.9	0.67	0.64
Unreconstructed gap with zero filling	74.6	0.99	0.70
Cubic interpolation	77.2	0.90	0.76
Cardio-Diffusion reconstruction	84.5	0.95	0.84

Figure 17 summarizes the classification metrics across the four strategies. The proposed Cardio-Diffusion reconstruction improves overall classification performance while maintaining high sensitivity for arrhythmic beats. Note: The original-signal baseline in this table corresponds to the same reconstruction-evaluation subset and classifier protocol used for comparison with gap-filling strategies. The secondary beat-level classifier reported in Section 3.9 uses a different random beat-level split and is therefore not directly comparable.

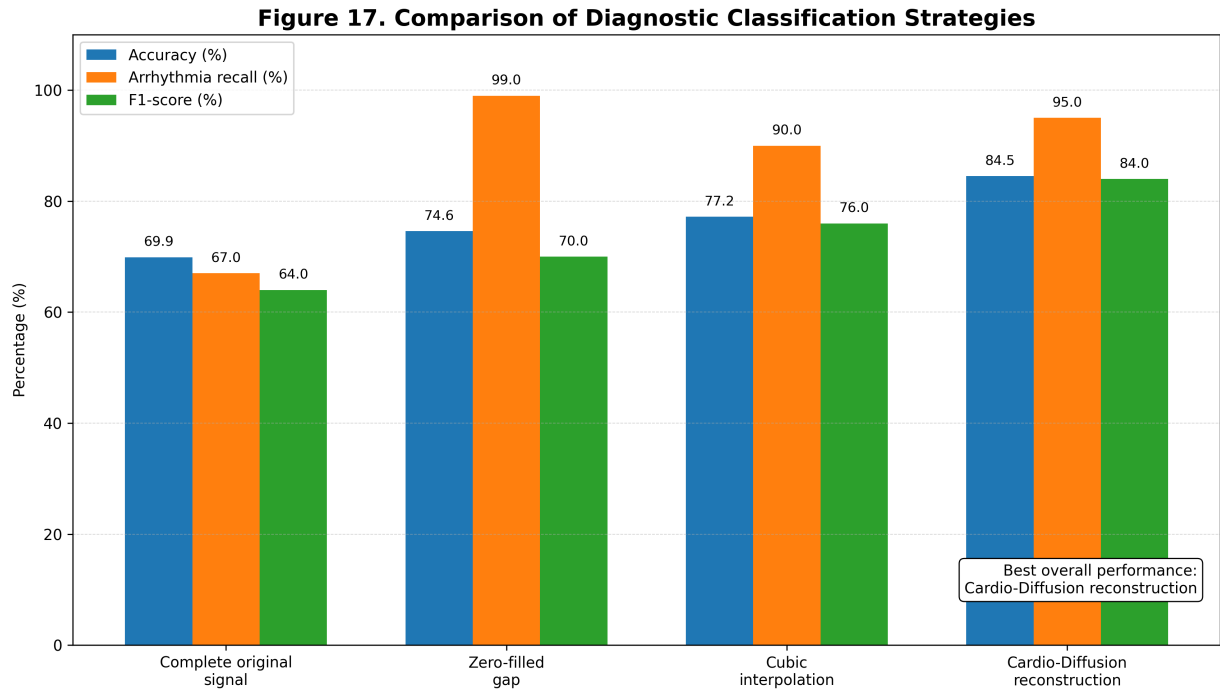


Figure 17. Comparison of diagnostic classification strategies. Cardio-Diffusion reconstruction improves classification performance compared with zero-filled gaps and cubic interpolation while maintaining high arrhythmia sensitivity.

3.7. Confusion Matrix of the Proposed System

Figure 18 presents the confusion matrix obtained by the classifier using reconstructed ECG segments. The test set contained 193 beats, including 70 Normal beats and 123 Arrhythmia beats. The classifier correctly classified 163 samples, corresponding to an overall accuracy of 84.5%.

The system correctly identified 117 of 123 arrhythmic samples, corresponding to an arrhythmia sensitivity of approximately 95.1%. It produced 24 false positives in the Normal class and 6 false negatives in the Arrhythmia class. This behavior is relevant for screening-oriented analysis, where false negatives are usually more critical than false positives.

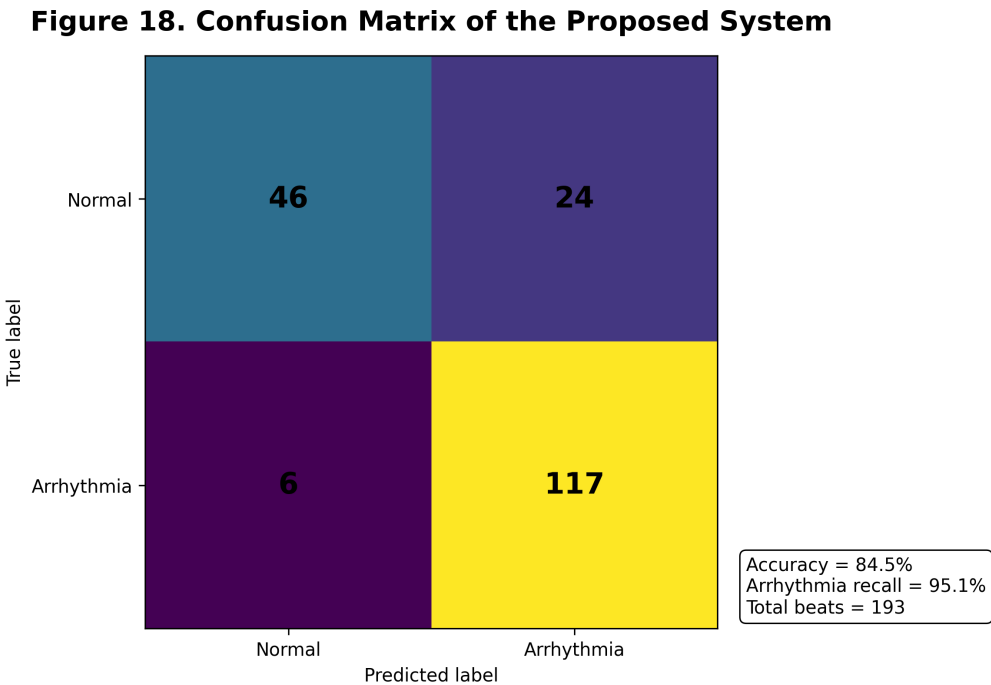


Figure 18. Confusion matrix for arrhythmia classification using reconstructed ECG segments. The classifier achieved an overall accuracy of 84.5% and correctly identified 117 of 123 arrhythmic samples.

3.8. Comparison with Related Methods

Table 9 places the proposed Cardio-Diffusion framework in the context of recent ECG reconstruction and diffusion-based signal imputation methods. Unlike most previous approaches, the proposed framework evaluates both signal-level reconstruction and downstream arrhythmia classification.

Table 9. Comparison of Cardio-Diffusion with recent ECG reconstruction and diffusion-based methods.

Study	Method	PRD (%)	Correlation	Includes classifier
Adib et al. (2023)	DDPM	15–25	N/R	No
Alcaraz and Strodthoff (2023)	SSSD-ECG (S4)	8–18	N/R	No
Neifar et al. (2024)	DiffECG	10–20	N/R	No
This work	Cardio-Diffusion	5.5–11.9	0.98	Yes, 84.5%

As shown in Table 9, Cardio-Diffusion achieves reconstruction metrics that are competitive with recent diffusion-based methods. More importantly, it integrates the reconstruction stage with a downstream diagnostic classifier, which is not commonly evaluated in previous ECG diffusion studies.

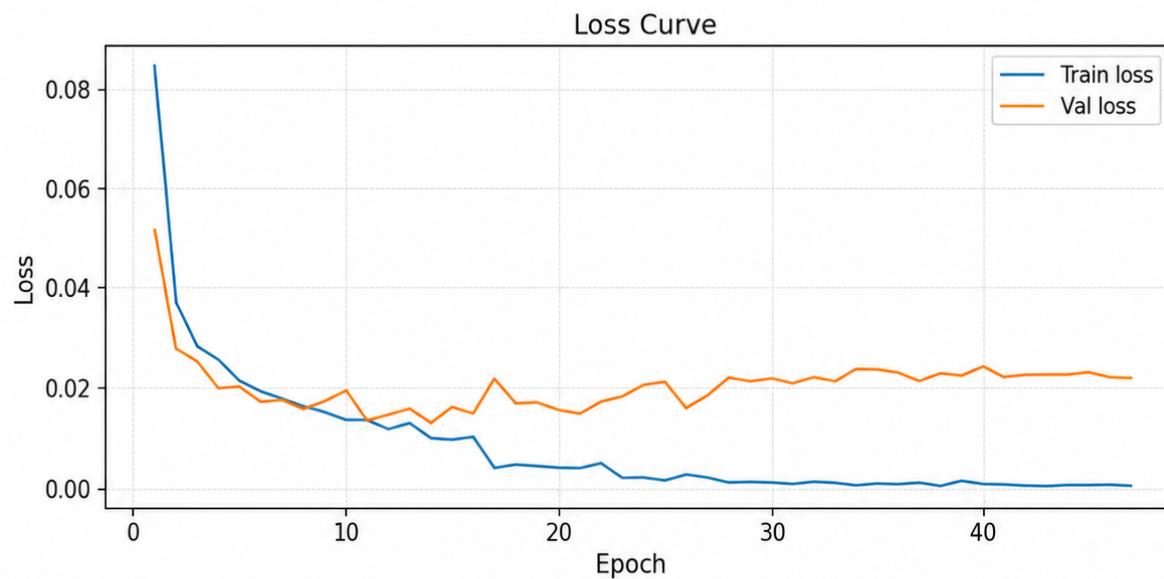
3.9. Secondary Beat-Level Classification on Original ECG Signals

As an additional experiment, a beat-level classifier was trained and evaluated using original ECG segments without generative reconstruction. The objective of this experiment

was to verify whether the classification architecture was capable of separating Normal and Arrhythmia morphologies when complete ECG information was available. Unlike the reconstruction-based diagnostic evaluation, this experiment used a random beat-level split; therefore, the result should be interpreted as a morphological separability benchmark rather than as patient-independent clinical validation.

The classifier achieved an overall accuracy of 98.99%, with a precision of 98.99%, recall of 98.01%, F1-score of 98.50%, and ROC-AUC of 99.81%. The best model was obtained at epoch 32, and the optimal decision threshold selected on the validation set was 0.62. The confusion matrix contained 2374 true Normal beats, 1184 true Arrhythmia beats, 12 false positives, and 24 false negatives.

(a) Training and validation loss



(b) Accuracy and F1-score

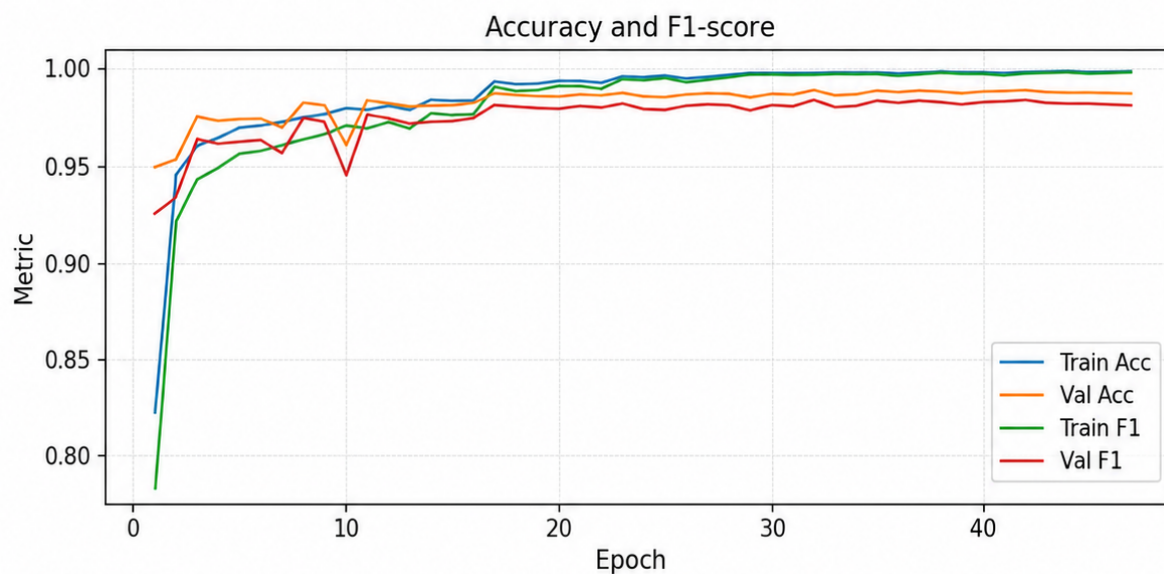


Figure 19. Training behavior of the secondary beat-level classifier trained on original ECG segments. Panel (a) shows the training and validation loss curves, while panel (b) shows the evolution of accuracy and F1-score.

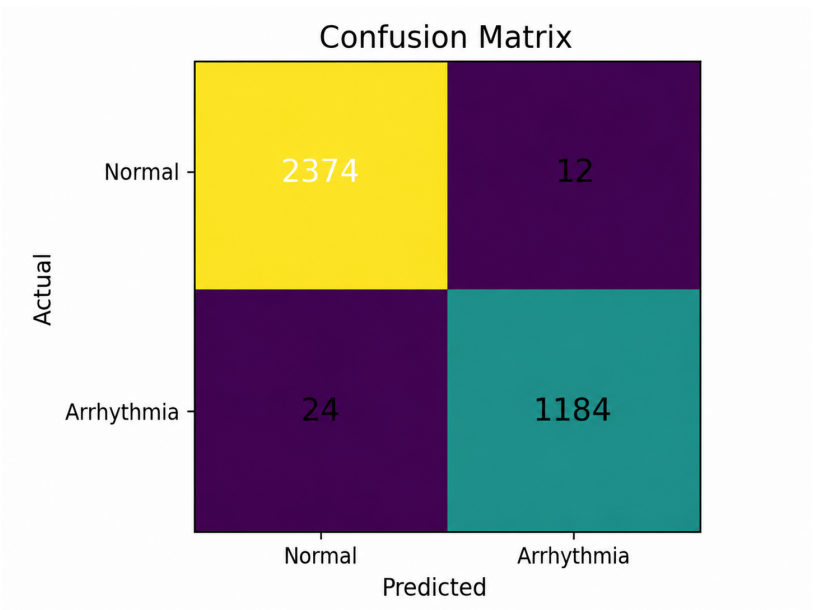


Figure 20. Confusion matrix of the secondary beat-level classifier trained on original ECG segments using a random beat-level split. The classifier achieved 98.99% accuracy, with 2374 true Normal beats, 1184 true Arrhythmia beats, 12 false positives, and 24 false negatives.

Table 10. Secondary beat-level classification results using original ECG signals and a random beat-level split.

Metric	Value
Input signal	Original ECG segments
Split mode	Random beat-level split
Sequence length	512 samples
Sampling frequency	360 Hz
Batch size	64
Learning rate	0.0008
Weight decay	0.0001
Best epoch	32
Decision threshold	0.62
Accuracy	98.99%
Precision	98.99%
Recall	98.01%
F1-score	98.50%
ROC-AUC	99.81%
Confusion matrix	$\begin{bmatrix} 2374 & 12 \\ 24 & 1184 \end{bmatrix}$

Although these results are substantially higher than the reconstruction-based classification results, they should not be interpreted as direct evidence of clinical generalization. Since the experiment used a random beat-level split, beats from the same record may appear in both training and testing subsets. Therefore, this experiment demonstrates that the classifier can separate ECG morphologies effectively under complete-signal conditions, but strict record-level or patient-level validation is still required for clinical claims.

3.10. Summary of Main Findings

The experimental results can be summarized in four main findings. First, Cardio-Diffusion V4 reconstructs most representative ECG segments with PRD values below 10%, although paced or highly aberrant morphologies remain challenging. Second, anatomical-region analysis confirms that the QRS complex is the most difficult region to reconstruct

because of its abrupt slopes and high-frequency morphology. Third, diffusion-based reconstruction outperforms cubic interpolation in the downstream classification experiment. Fourth, the classifier applied to reconstructed ECG segments achieves high arrhythmia sensitivity, correctly identifying 117 of 123 arrhythmic samples.

Overall, these results support the feasibility of using conditional diffusion models to reconstruct missing ECG segments while preserving information relevant for arrhythmia classification.

4. Discussion

The results presented in this study show that Cardio-Diffusion can reconstruct missing ECG segments with competitive signal fidelity while preserving information useful for downstream rhythm analysis. The experimental evidence must be interpreted through two complementary levels. The first level corresponds to the main reconstruction-based pipeline, where ECG gaps are artificially introduced, reconstructed by the conditional diffusion model, and then evaluated using both signal-level metrics and a binary classifier. The second level corresponds to a secondary beat-level classification experiment performed on original ECG segments, without reconstruction, using a random beat-level split. These two evaluations address different questions: the first measures the usefulness of Cardio-Diffusion as a signal-rescue method, while the second measures the morphological separability of Normal and Arrhythmia beats under complete-signal conditions.

4.1. Interpretation of the Reconstruction Results

Cardio-Diffusion V4 achieved an average PRD of 13.27% across the five representative MIT-BIH records evaluated in the reconstruction experiment. This mean value was strongly affected by record 234, which reached a PRD of 37.53% due to its highly aberrant paced morphology. In contrast, the remaining records achieved substantially lower errors, with PRD values of 5.45% for record 222, 5.81% for record 107, 7.93% for record 101, and 9.61% for record 118. These results indicate that the model reconstructs missing ECG regions accurately when the hidden morphology is compatible with the visible context and with the patterns learned during training.

The anatomical-region analysis confirms that reconstruction difficulty is not uniform across the cardiac cycle. The ST segment obtained the lowest PRD, with 6.38%, while the QRS complex obtained the highest PRD, with 11.88%. This behavior is physiologically consistent. The ST segment is smoother and closer to the isoelectric line, whereas the QRS complex contains abrupt slopes, narrow peaks, and high-frequency components. Therefore, small temporal or amplitude deviations inside the QRS complex can produce a larger error than similar deviations in smoother regions. This result supports the need to evaluate ECG reconstruction by anatomical waveform component rather than relying only on a global metric.

The comparison against cubic interpolation further shows the advantage of a morphology-aware generative model. Interpolation methods can maintain mathematical continuity between the edges of a missing interval, but they do not learn the physiological structure of the ECG waveform. As a result, they tend to oversmooth abrupt QRS morphology and underestimate sharp transitions. Cardio-Diffusion, by contrast, uses a learned generative prior conditioned on the visible ECG context and the binary mask, allowing it to reconstruct hidden intervals with greater morphological consistency.

4.2. Diagnostic Utility of Reconstructed ECG Segments

The reconstruction-based diagnostic evaluation showed that ECG segments reconstructed by Cardio-Diffusion preserve information useful for binary arrhythmia classifica-

tion. In this setting, the classifier applied to reconstructed ECG segments achieved 84.5% accuracy, 0.95 arrhythmia recall, and 0.84 F1-score. The confusion matrix showed that the system correctly identified 117 of 123 arrhythmic samples, corresponding to an arrhythmia sensitivity of approximately 95.1%.

This high arrhythmia sensitivity is relevant in a screening-oriented Holter scenario, where false negatives are usually more critical than false positives. A system that misses arrhythmic events could delay diagnosis, whereas false positives can still be reviewed by a clinician. Nevertheless, the system also produced 24 false positives in the Normal class, indicating that the reconstruction-based classifier favors sensitivity over specificity. This behavior is acceptable for preliminary screening but would require calibration, external validation, and clinical review before practical deployment.

The 84.5% result should be interpreted as the diagnostic performance of the complete reconstruction-based pipeline, not as the maximum classification capacity of the dataset. Its purpose is to evaluate whether ECG segments remain diagnostically useful after artificial signal loss and diffusion-based reconstruction. Therefore, this result is directly linked to the central objective of Cardio-Diffusion: rescuing incomplete ECG segments that would otherwise lose diagnostic value.

4.3. Secondary Beat-Level Classification on Original ECG Signals

A separate secondary experiment was performed using original ECG beats without generative reconstruction. This experiment evaluated the classification architecture under complete-signal conditions and measured the morphological separability between Normal and Arrhythmia beats in the MIT-BIH domain. The classifier achieved 98.99% accuracy, 98.99% precision, 98.01% recall, 98.50% F1-score, and 99.81% ROC-AUC. The best model was obtained at epoch 32, and the validation-selected decision threshold was 0.62. The confusion matrix contained 2374 true Normal beats, 1184 true Arrhythmia beats, 12 false positives, and 24 false negatives.

These results demonstrate that the selected classifier architecture can strongly separate Normal and Arrhythmia morphologies when complete original ECG information is available. However, this experiment used a random beat-level split. In ECG analysis, random beat-level partitioning can overestimate real-world performance because beats from the same record or patient may appear in both training and testing subsets. Therefore, the 98.99% accuracy should be reported as a secondary morphological benchmark rather than as evidence of patient-independent clinical generalization.

This distinction is essential for a fair interpretation of the results. The reconstruction-based evaluation measures the behavior of the complete Cardio-Diffusion pipeline under missing-signal conditions. The beat-level experiment measures the discriminative capacity of the classifier when no signal region is missing. Consequently, the 98.99% result strengthens the study by showing that the classifier is capable of high morphological discrimination, but it should not replace the reconstruction-based result as the principal diagnostic evidence of the proposed framework.

4.4. Relationship between the 84.5% and 98.99% Classification Results

The two classification results are not contradictory because they correspond to different experimental conditions. The 84.5% accuracy was obtained in the reconstruction-based pipeline, where ECG regions were artificially removed and then reconstructed before classification. This experiment is more directly related to the central objective of rescuing incomplete ECG segments.

The 98.99% accuracy was obtained using original ECG beats and a random beat-level split. This experiment is useful as an upper-bound reference for morphology separability

inside the MIT-BIH domain, but it is less strict from the perspective of patient-independent validation. Therefore, both results should be presented as complementary: Cardio-Diffusion demonstrates that reconstructed ECG remains diagnostically useful, while the secondary classifier demonstrates that the architecture can achieve very high performance when complete ECG morphology is available.

4.5. Training Convergence and Computational Feasibility

The training convergence curve indicates a stable learning process across the accumulated development stages. The validation loss decreased from approximately 2.670 at the start of training to a minimum of 0.526 at accumulated epoch 820, corresponding to an overall reduction of approximately 80%. The selected V4 checkpoint at accumulated epoch 1096 did not correspond to the absolute minimum validation loss, but it was chosen because it provided a better balance between validation behavior and visually coherent ECG reconstruction.

This selection criterion is important in generative biomedical modeling. The lowest numerical loss does not always correspond to the most useful reconstruction, especially when the model is evaluated visually and diagnostically. For this reason, checkpoint selection should consider quantitative validation trends, morphological stability, and the absence of visually disruptive artifacts.

A relevant engineering aspect of this work is that Cardio-Diffusion was trained and evaluated on consumer-grade hardware. The use of an NVIDIA GeForce RTX 4050 GPU with 6 GB of VRAM demonstrates that diffusion-based ECG reconstruction can be developed in academic environments without enterprise-level GPU infrastructure. Although the accumulated training time remains considerable, this computational feasibility is relevant for institutions with limited resources.

4.6. Critical Comparison with Related Work

The literature reviewed in Table 1 shows a clear evolution from classical ECG filtering and artifact detection toward deep learning classification and generative reconstruction. Classical artifact detection systems are useful for identifying low-quality segments, but they usually discard the affected samples instead of reconstructing them. Cardio-Diffusion changes this paradigm by attempting to recover the missing segment and preserve its diagnostic usefulness.

Classification-only methods such as 1D CNNs, ResNet-based architectures, or recurrence-plot approaches can achieve strong performance when the ECG window is complete. However, they do not directly solve the problem of missing signal intervals. If the diagnostically relevant QRS morphology is partially or completely lost, a classifier alone cannot recover the missing information. The proposed pipeline addresses this limitation by placing a generative reconstruction stage before classification.

GAN-based ECG generation methods have been explored for augmentation and synthesis, but they are commonly affected by training instability and mode collapse. Diffusion models provide a more stable alternative because they learn a gradual denoising process. Cardio-Diffusion follows this direction and adapts the DDPM framework to local ECG gap reconstruction using a binary mask and a 1D U-Net architecture.

Compared with generic time-series imputation models such as CSDI, the present approach is more specialized for ECG morphology. The model uses two ECG leads, explicit mask conditioning, and a hybrid loss designed to preserve time-domain, frequency-domain, and slope-related information. Compared with more complex diffusion models that incorporate structured state-space blocks or multi-lead reconstruction, Cardio-Diffusion offers a

simpler and more accessible implementation while still evaluating downstream diagnostic usefulness.

Table 11 summarizes the qualitative position of Cardio-Diffusion relative to representative related methods.

Table 11. Qualitative comparison between Cardio-Diffusion and representative related methods.

Study	Main task	Generative method	Main advantage of Cardio-Diffusion
Adib et al. (2023)	ECG beat synthesis	DDPM	Performs contextual gap reconstruction and evaluates classification utility
Alcaraz and Strodthoff (2023)	ECG imputation	Diffusion + S4	Uses a simpler ECG-specific pipeline with downstream classifier evaluation
Jenkins et al. (2024)	Template-guided imputation	DDPM + physiological template	Does not depend on an external patient-specific template
Cardoso et al. (2024)	Multi-lead reconstruction	DDM + Bayesian inverse reconstruction	Focuses on local missing-segment reconstruction and diagnostic preservation
Bedin et al. (2025)	12-lead ECG synthesis	Diffusion + S4	Prioritizes local QRS/ST-T gap imputation instead of full-lead synthesis
Kranz et al. (2025)	Long-duration ECG generation	Latent diffusion	Evaluates reconstructed windows using a binary arrhythmia classifier
This work	ECG imputation + classification	DDPM + 1D U-Net	Integrated pipeline from missing ECG reconstruction to arrhythmia classification

4.7. Clinical and Engineering Implications

From a clinical perspective, the results suggest that incomplete ECG segments should not necessarily be discarded automatically. If the missing interval can be reconstructed with sufficient fidelity and clearly marked as generated, the segment may still provide useful information for automated screening. This is particularly relevant in ambulatory Holter monitoring, where corrupted segments are common due to movement, sweating, electrode detachment, and cable instability.

From an engineering perspective, the proposed framework demonstrates that conditional diffusion can be adapted to one-dimensional ECG signals using a mask-guided input representation. The binary mask explicitly informs the model which samples are known and which samples must be reconstructed. This design makes the system flexible because different gap sizes and anatomical locations can be evaluated without changing the entire architecture.

However, the reconstructed signal should never be treated as a direct replacement for the measured ECG. A generated segment is an estimate conditioned on surrounding context, not a clinical measurement. Therefore, in a real deployment scenario, reconstructed regions should be visually highlighted and associated with confidence or uncertainty scores.

5. Limitations

Although the results are promising, the present study has several limitations that must be considered before interpreting Cardio-Diffusion as a clinically deployable system.

- Dataset size and diversity.** The model was trained and evaluated primarily on the MIT-BIH Arrhythmia Database. Although MIT-BIH is a standard benchmark, it contains only 48 records and does not fully represent the diversity of modern Holter recordings, wearable devices, pediatric patients, elderly patients, acquisition systems, and real-world noise conditions. This limitation is especially relevant for

- rare morphologies such as paced beats, as shown by the high PRD obtained for record 234.
- Artificial masking of missing data.** The missing regions used in the experiments were simulated using artificial binary masks. This approach enables quantitative comparison against ground truth, but real Holter signal loss may include gradual transitions, intermittent contact noise, electrode motion artifacts, saturation, and overlapping noise before or after the missing interval. Therefore, artificial masking does not fully reproduce the complexity of real-world ECG corruption.
 - Two-lead input representation.** The system was designed for two ECG leads, typically MLI and V1 in MIT-BIH. In clinical practice, arrhythmia interpretation often benefits from 12-lead ECG information. Extending the architecture to 12 leads could improve reconstruction by using spatial information from other leads, but it would also require architectural changes and greater memory capacity.
 - The reconstruction-based classifier did not reach 90% accuracy.** The classifier evaluated on reconstructed ECG segments achieved 84.5% accuracy. This is below the initial 90% target, but it should be interpreted in the context of missing-signal reconstruction. When a large gap covers the QRS complex, some of the most discriminative information for arrhythmia classification may be lost. The generative model can reconstruct a plausible waveform based on context, but it cannot guarantee recovery of diagnostic information that is absent from the visible samples.
 - Beat-level random split may overestimate classification performance.** The secondary classifier achieved 98.99% accuracy on original ECG signals, but this experiment used a random beat-level split. Beats from the same record may therefore appear in both training and testing subsets, which can inflate performance due to patient-specific or record-specific morphology. This result should be interpreted as morphological separability, not as patient-independent clinical validation.
 - Stochastic variability of diffusion reconstructions.** Diffusion-based reconstruction is inherently stochastic. Two reconstructions of the same corrupted segment may differ slightly, especially when the hidden region contains complex morphology. Repeated sampling or ensemble strategies can reduce this variability, but they also increase computational cost.
 - Computational validation only.** The present evaluation was performed in silico using benchmark ECG data. No prospective clinical validation was conducted with real Holter recordings from patients under ambulatory conditions. Therefore, the current results should be interpreted as proof of concept rather than clinical validation.

Table 12 summarizes these limitations and the corresponding improvement strategies.

6. Recommendations and Future Work

Based on the limitations identified above, several development lines are recommended for future versions of Cardio-Diffusion.

- Transfer learning with larger ECG datasets.** A model pretrained on larger datasets such as MIMIC-IV could learn a broader distribution of ECG morphologies and then be fine-tuned on MIT-BIH or patient-specific data. This strategy may improve reconstruction of rare or highly aberrant morphologies, such as paced beats.
- Strict patient-independent validation.** Future experiments must use record-level, patient-level, and cross-database splits, especially for classification. This is necessary to determine whether the model generalizes beyond patient-specific morphology and recording-specific patterns.
- Extension to 12-lead ECG reconstruction.** The 1D U-Net architecture could be extended to accept 12 simultaneous leads. This would allow the reconstruction of a

Table 12. Main limitations of the present study and proposed improvement strategies.

Limitation	Impact	Recommended improvement
Single main dataset	Limited generalization to other devices, patients, and acquisition settings	Evaluate on PTB-XL, INCART, QTDB, European ST-T, MIMIC-IV, and real Holter data
Artificial masks	Do not fully represent real signal loss	Use realistic artifact masks from documented Holter failures
Two-lead input	Does not exploit full clinical ECG spatial information	Extend the architecture to multi-lead or 12-lead reconstruction
Reconstruction-based accuracy below 90%	Indicates that some diagnostic information remains difficult to recover after QRS loss	Use longer rhythm segments and stronger classifiers
Beat-level random split	May overestimate classification performance	Validate with strict record-level, patient-level, and cross-database splits
Stochastic reconstruction	Introduces uncertainty in reconstructed morphology	Use uncertainty estimation, repeated sampling, and confidence scores
Computational-only validation	Does not prove clinical effectiveness	Conduct prospective validation with real Holter recordings

- missing interval in one lead to benefit from information preserved in other leads, potentially improving morphology consistency.
- Classification using longer rhythm segments.** Instead of classifying isolated beats only, future work should evaluate longer rhythm segments of 10–30 seconds. In a longer segment, the loss of one QRS complex has less impact on the diagnostic decision because rhythm regularity, heart rate, and repeated beat patterns remain available.
 - Incorporation of clinical multimodal information.** Patient metadata such as age, sex, medication, and clinical history could be encoded as additional conditioning variables. This may improve diagnostic classification and reconstruction consistency in patient-specific contexts.
 - Optimization for real-time inference.** Future versions should reduce inference time using DDIM sampling, diffusion distillation, pruning, quantization, or lightweight architectures. This is necessary for long-duration Holter processing or near-real-time monitoring.
 - Prospective clinical validation.** Before clinical use, Cardio-Diffusion should be evaluated in a prospective study with real Holter recordings. The study should compare diagnostic yield, false alarm rate, and number of non-diagnostic studies with and without generative reconstruction.
 - Classification-aware reconstruction metrics.** PRD and correlation measure signal similarity, but they do not always predict classification usefulness. Future work should investigate reconstruction metrics aligned with downstream arrhythmia detection.
 - Robustness against realistic missing-data patterns.** Future experiments should include realistic missing patterns based on documented Holter failures, including gradual electrode detachment, intermittent contact, saturation, and motion artifacts.
- Table 13 summarizes the recommended development roadmap.

7. Conclusions

This work presented **Cardio-Diffusion**, an integrated framework for reconstructing missing ECG segments using conditional denoising diffusion models and evaluating their usefulness for arrhythmia analysis. The proposed system combines ECG preprocessing, arti-

Table 13. Recommended roadmap for future development of Cardio-Diffusion.

Priority	Recommendation	Expected benefit
High	Validate with strict record-level, patient-level, and cross-database splits	Stronger generalization evidence
High	Pretrain or fine-tune using larger datasets such as MIMIC-IV	Better reconstruction of rare morphologies
High	Evaluate real Holter artifact patterns	More realistic clinical assessment
Medium	Extend the system to 12 ECG leads	Improved spatial consistency
Medium	Classify longer rhythm segments	Better robustness when QRS information is missing
Medium	Add uncertainty estimation	Safer interpretation of reconstructed regions
Medium	Optimize inference speed	Practical long-record processing
Low	Add clinical metadata conditioning	More personalized reconstruction and classification

ificial missing-segment simulation, a conditional DDPM based on a 1D U-Net with attention, anatomical reconstruction analysis, and binary Normal-versus-Arrhythmia classification.

The experimental results show that diffusion-based reconstruction is a feasible strategy for recovering missing ECG morphology under controlled masking conditions. Cardio-Diffusion V4 achieved a PRD of 6.38% in the ST segment and 11.88% in the QRS complex, confirming that smoother ECG regions are reconstructed with lower error, while abrupt high-frequency regions remain more challenging. The average PRD across the five representative MIT-BIH records was 13.27%, although this value was strongly affected by record 234, which contained highly aberrant paced morphology and produced the highest reconstruction error.

The reconstruction-based diagnostic evaluation demonstrated that reconstructed ECG segments preserve clinically relevant information for arrhythmia screening. The classifier applied to Cardio-Diffusion reconstructions achieved 84.5% accuracy and approximately 95.1% arrhythmia sensitivity, correctly identifying 117 of 123 arrhythmic samples. Although this result did not reach the initial 90% accuracy target, it confirms that the reconstructed signals retain diagnostic value after artificial signal loss and diffusion-based recovery.

As a complementary experiment, a secondary beat-level classifier trained on original ECG signals achieved 98.99% accuracy, 98.99% precision, 98.01% recall, 98.50% F1-score, and 99.81% ROC-AUC. This result confirms that the selected classification architecture can strongly separate Normal and Arrhythmia morphologies when complete ECG information is available. However, because this experiment used a random beat-level split, it should be interpreted as a morphological separability benchmark rather than as patient-independent clinical validation.

A central contribution of this study is the connection between generative ECG reconstruction and downstream diagnostic evaluation. Unlike approaches that only synthesize ECG signals or only classify clean ECG windows, Cardio-Diffusion evaluates whether reconstructed ECG segments remain useful for arrhythmia analysis. This makes the framework a proof of concept for rescuing incomplete ECG segments that would otherwise be discarded in automated Holter processing.

The study also identifies important limits of generative ECG imputation. When the missing interval covers highly abnormal or paced QRS morphology, the surrounding context may not contain enough information to recover the exact diagnostic waveform. Therefore, Cardio-Diffusion should not be interpreted as a replacement for the original ECG signal or for expert clinical interpretation. Reconstructed regions must remain clearly

identified, and future implementations should include uncertainty estimation or confidence scoring.

From a computational perspective, the model was developed using consumer-grade GPU hardware, showing that diffusion-based ECG reconstruction can be explored in academic or resource-limited environments. Nevertheless, the accumulated training time and the iterative nature of diffusion sampling remain important practical challenges. Future versions should optimize inference speed through DDIM acceleration, model distillation, pruning, quantization, or lightweight latent diffusion strategies.

In summary, Cardio-Diffusion establishes a solid experimental basis for using conditional diffusion models as a signal-rescue strategy for incomplete ambulatory ECG recordings. The framework achieved competitive reconstruction quality, preserved information useful for arrhythmia classification, and demonstrated strong secondary beat-level classification performance on original ECG signals. Future work should focus on larger and more diverse datasets, real Holter artifact patterns, strict patient-independent and cross-database validation, extension to 12-lead ECG, longer rhythm-segment classification, uncertainty-aware reconstruction, and prospective clinical assessment.

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